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DRILLED PIER FOUNDATIONS

PURPOSE

This document establishes guidelines and recommends procedures for design of drilled pier foundations.

A drilled pier is also referred to as a drilled shaft, belled pier, and under-ream foundation. This document is not intended for auger cast piles.

SCOPE

This document includes the following:

- Feasible areas of application of drilled piers.
- Minimum requirements from a Geotechnical Report and points out the important role of a Geotechnical Consultant.
- Suggested and recommended pier details.
- Methods for computing axial and lateral capacities of drilled piers.
- Brief discussion of settlement, group efficiencies, special requirements in high seismic areas, and potential pier buckling.
- Sample computations.

APPLICATION

Drilled piers should be used when recommended by the Geotechnical Consultant for a particular site or when they are the most economical of several alternatives allowed by the Geotechnical Consultant. The Lead Structural Engineer will decide when and in which areas drilled piers will be used for a specific Project.

GEOTECHNICAL CONSULTANT

The design of drilled piers is a very complex subject. Successful design of deep drilled piers requires close collaboration with the Geotechnical Consultant.

Report

The Geotechnical Report should address or provide the following:

- Feasibility of using drilled piers.
- Suitability of soil for drilling and a subsurface soil profile.

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- Recommended methods of construction and a discussion on potential problems.
- Suitability for underreams and recommended bell to shaft diameter ratios.
- Suitable diameter of piers.
- Suitable depths of straight piers and underreams.
- Axial capacities, both compression and uplift, for suitable diameters and lengths. These axial capacities should list separately end bearing and skin friction capacities and factor of safety used. Estimated settlements for these capacities, spacing, and group efficiency factors will be given. Group effects of adjacent foundations will be addressed. Methods to compute axial capacities will be given.
- Lateral capacities for different pier diameters, depths, and eccentricities such as moments at top of shaft. Group effects can significantly reduce allowable lateral capacity. The report will provide recommended spacing and group effect factors. The soil lateral capacity will be given for a maximum allowable deflection of 0.25 inches at working loads at the top of pier. Deflection at ultimate soil resistance will be greater.
- Recommendations for lateral load analysis using nonlinear computer program and P-y curves if applicable.
- Modulus of lateral subgrade reaction, k_h.
- Soil modulus (E_s), density (γ), Poisson's ratio (μ), 50 percent strain ($\varepsilon 50$), and undrained shear strengths (C_u) for clays.
- Adhesion coefficient (α_z) for cohesive soils for computation of skin friction if applicable.
- Angle of internal friction for sand and pier (ϕ) for cohesionless soil for computation of skin friction.
- For cohesionless soils, uncorrected field values on standard penetration tests, *N*.

GENERAL

Single piers may be used to support individual columns such as pipe racks, structures, and T supports. In high seismic areas, UBC (Uniform Building Code) pier cap interconnection requirements will be reviewed.

A group of piers with a pier cap or grade beams may be used to support structures or vertical and horizontal vessels similar to driven piers.

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Advantages

- Drilled piers are economical since no forming and very little excavation and backfill are required.
- Drilled piers can be designed and constructed to support large axial and lateral loads. The diameter can be large and the pier extended to great depths requiring fewer piers. Pier caps may be eliminated.
- Drilled piers can carry large loads with minimal settlement.
- Larger diameter drilled piers allow direct inspection of bearing area and soil at base.
- Drilled piers eliminate much of the vibration and noise associated with pier driving affecting nearby existing installations.
- Drilled piers can go through a boulder soil where driven piers might be deflected. Boulders can be removed directly or broken with special tools and then removed using temporary casing.

Disadvantages

- Unsuitable soil may cave in during underream operation.
- Bad weather conditions may make drilling and concreting difficult.
- Need to dispose spoils from drilling, belling, and slurry operations. In environmental sensitive areas, cost may be prohibitive due to local regulations.
- Slurry method of construction in an existing plant area is messy and increases cleanup costs.
- For large diameter piers in weak soils, there may be ground loss in the vicinity of the drilled hole if adequate precautions are not taken.
- Drilled piers in groups require special sequencing for drilling and concreting and may result in additional costs.
- Successful completion depends on drilling operator skills.
- Because of their larger diameters, spacing will generally be greater than for driven piers resulting in larger pier caps when used in groups.

PIER DETAILS

- Shaft diameters will be sized to agree with equipment available to the Drilling Contractors in the vicinity of the jobsite that are likely to be doing the work.

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- On a specific site, the number of different shaft diameters will be reduced to a minimum to speed construction.

- Bells will be sized in 6 inch increments. The bell to shaft diameter ratio will be a maximum of 3. Quite often soil conditions will restrict this ratio to 2.

- The minimum toe height of the bell will be 6 inches.

- According to ACI (American Concrete Institute) 336.3R – 93(reapproved 2006), for shafts larger than 30 inches, the minimum slope of the side of the bell will be 55 degrees.

- Use of this criteria requires a 60 series underreamer. A 60 series underreamer for shafts larger than 30 inches does not fit on a rig without jacking and is not preferred by contractors. Also, the time required to form a 60 degree bell is longer; therefore, the cost is increased. Use of 45 degree bell slope is permitted provided concrete stress in unreinforced bell is less than allowable for plain concrete and provided the soil can stand the flatter slope without caving. Use of 45 degree slopes requires approval by the Lead Structural Engineer. Details of stresses in plain underreams are found in Plain Concrete Underream for Drilled Shafts, by J.S. Farr, L.C. Reese, ASCE (American Society of Civil Engineers) Structural Journal, June 1980.

- The reinforcement will preferably extend full length of the shaft. For very deep shafts without tension, reinforcement may not be necessary for lower section.

- Minimum clear distance of vertical reinforcement will be 3 inches.

- Minimum cover to vertical reinforcement is 3 inches and 4 inches in cased piers where casing is to be withdrawn.

- At pier caps, the shaft will extend a minimum of 4 inches into the cap. The shaft is considered free headed for this condition.

- Pipe cap edge distance from the shaft will be at least 6 inches.

- For high seismic areas , the upper part of shaft and the cap will be subjected to high moments and shear. The reinforcement splices at this interface and the cap reinforcement require special spiral and hoop reinforcement. For additional details, refer to ACI 318.

LOADING

Drilled piers will be designed for axial and lateral loads, including group effects.

Static

The design procedure presented herein is for static loading and repeated loading. Live and wind loads are usually considered as static equivalents in analysis.

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Seismic

In high seismic areas, cyclic loading has a severe effect on soil behavior. In particular, the lateral capacity could be drastically reduced. The lateral subgrade modulus could be reduced to 30 percent of the initial value. Cyclic load combined with group action may reduce the initial lateral subgrade modulus to 10 percent of its value. Close collaboration with the Geotechnical Engineer is required in high seismic areas. Field tests may also be required.

COMPUTER DESIGN

Soil pier interaction is nonlinear and semi-empirical. For sites with high wind speeds or high seismic areas, some structures may require use of very deep drilled piers. These structures may require use of nonlinear computer programs. Use of computer programs, although easy to use, should be with close collaboration with the Geotechnical Engineer. The Lead Structural Engineer will decide which structures require detailed computer analysis.

Note: *A thorough understanding of the computer programs and method used is essential. The Geotechnical Engineer should provide all the necessary data and review results in some cases.*

NOTATIONS

A_b	Base area.
B_b	Bell or underream diameter.
C_u	Cohesive soils - cohesion determined from undrained triaxial, direct shear, or vane tests. (This is also called unconfined compressive strength or undrained shear strength).
C_{uz}	Cohesion at depth z.
C_{ub}	Cohesion 1 to 2 bell diameter below base.
D	Pier or shaft diameter.
E_p	Modulus of elasticity of pier material.
E_s	Modulus of elasticity of soil.
F_r	Reduction factor for bearing resistance for large bells.
I_P	Moment of inertia of pier section.
K_P	Cohesionless soils - rankine coefficient of passive lateral earth pressure.
L	Embedment depth such as from grade to bottom of pier or bell.
M_{max}	Unfactored moment capacity of pier section. This should be obtained from ultimate pier section capacity.

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M_{maxn}	Unfactored maximum negative moment in a short fixed headed pier.
M_{maxp}	Unfactored maximum positive moment in a short or intermediate fixed headed pier.
N	Cohesionless soil - standard penetration number.
P	Actual lateral load at top of pier.
P_{ULT}	Ultimate lateral load at top of pier at which soil failure occurs. This is ultimate soil resistance. It is not ultimate load for concrete design.
P_a	Allowable (service load) lateral load at top of pier.
Q_B	Ultimate soil compression capacity in base resistance.
Q_S	Ultimate soil side or skin resistance (friction).
Q_{TC}	Ultimate soil compression capacity of drilled pier foundation. This is total of side and base resistance.
Q_{TT}	Ultimate soil tension capacity of drilled pier foundation. This is a total of side and base resistance.
Q_{BELL}	Ultimate bell pull out capacity.
S_C	Cohesive soils - shape factor for end bearing.
W_C	Weight of concrete and soil above bell.
Y_O	Deflection at grade at allowable or actual applied load.
a	Used in computation of base resistance reduction due to large bell diameter.
b	Used in computation of base resistance reduction due to large bell diameter.
d_c	Cohesive soil - depth factor for end bearing.
e	Lateral load eccentricity (Refer to Attachment 04).
f_{sz}	Ultimate load transfer in skin friction at depth z.
k	Cohesive soils - modulus of subgrade reaction.
k_h	Cohesionless soils - coefficient of horizontal subgrade reaction.
n_h	Cohesionless soil - coefficient of lateral subgrade reaction for a long pier with a width of unity at a depth of unity. This coefficient is independent of pier length and stiffness. It depends on soil relative density and ground water table.
q	Surcharge such as soil density times depth.
q_b	Ultimate soil end bearing stress.
q_{br}	Reduced ultimate soil end bearing stress. This reduction is due to large bell diameter.
z	Depth below ground surface.
α	Cohesive soil - for Alpha method of skin friction, this is an empirical factor that varies with depth. Refer to Attachment 14.

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β	Cohesive soil - a parameter to compute dimensionless length.
	Cohesionless soil - a factor in computation of side resistance using Beta method.
γ	Soil density.
φ	Soil angle of internal friction.
λ	Cohesionless soil - a parameter to compute dimensionless length.
σ_z	Cohesionless soil - vertical effective stress at depth z. (This is soil density times depth).
μ	Poisson's ratio of soil.
ε_{50}	50 percent strain of soil.

AXIAL COMPRESSION LOAD

- Allowable axial compression load strongly depends on allowable settlements.
- Short term settlements should be restricted to 1 inch. For some piping systems, it may be necessary to restrict short term settlement to 1/2 of an inch.
- Methods presented here are based on computation of ultimate soil resistance and application of safety factors. For good, dependable soil data, the factor of safety will be 2.5 to 3 for compression load.
- The general equation for computing the ultimate compression capacity of a drilled pier is:

$$Q_{TC} = Q_B + Q_S$$

where

Q_{TC} = Ultimate compression capacity of drilled pier foundation.
Q_B = Ultimate capacity of the base resistance.
Q_S = Ultimate capacity in side or skin resistance.

- For short rigid piers, a substantial portion of the load will be resisted at base.
- For intermediate and long piers, a substantial portion of load will be resisted by skin friction or side resistance. Usually, at a relatively small downward movement, the full skin friction resistance is mobilized. For piers in clays, a vertical movement on the order of 0.25 inches will mobilize full skin resistance. In sands, vertical movement on the order of 0.4 inches will mobilize full skin resistance.

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- The amount of displacement necessary to mobilize full base resistance is a function of the base diameter and is on the order of 10 percent of the base diameter. Settlements may be too excessive for development of full base resistance.

Cohesive Soils (Clay)

Side Resistance (Skin Friction)

- For skin friction resistance, the Alpha (α) method is used below.

$$f_{sz} = \alpha C_{uz} \leq 5.5\ ksf$$

where

f_{sz} = Ultimate load transfer in skin friction at depth *z*, *ks*
C_{uz} = Undrained shear strength at depth *Z*, *ksf*
α = Empirical factor that varies with depth

Refer to Attachment 14.

$$Q_s = \int_0^L f_{sz}\, da$$

where

dA = Differential area of the perimeter, square feet.

L = Embedment depth below ground surface, feet.

- For lateral deflections greater than 0.2 inches, the skin friction down to the point of zero deflection is neglected.

Base Resistance (End Bearing)

- For end bearing resistance, the general bearing capacity equation is used here.

$$q_b = 5.14\, C_{ub}\, S_c\, d_c + q$$

where

q_b = Ultimate soil end bearing stress, *ksf*

C_{ub} = Average undrained shear strength 1 to 2 bell diameters below base, *ksf*.

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S_c = Shape factor = 1.2 for circular base.

d_c = Depth factor = 1 + 0.2L/B_b; but not greater than 1.5.

L = Embedment depth of pier, feet.

B_b = Bell diameter, feet.

q = Total unit surcharge (Soil density x L), *ksf*.

- For a pier at least 2.5 bell diameters below grade and using net capacities (weight of drilled pier not considered as load).

q_b = 9 $C_{ub} \leq 80$ *ksf*

- q_b will be reduced when the base diameter B_b is greater than 75 inches to reduce excessive short term settlements.

Reduced $q_{br} = F_r\, q_b$

where

$$F_r = \left[\frac{2.5}{12a\ B_b + 2.5\, b}\right] \qquad F_r \leq 1.0$$

a = 0.0071 + 0.0021 (L/B_b) $\qquad a \leq 0.015$

b = 0.45 $\qquad 0.5 \leq b \leq 1.5$

Note: *Above reduction equation is from tests in very stiff clays, soft clay shales, and restricts q_{br} to a settlement of 2.5 inches. If end bearing constitutes more than 50 percent of design load, a minimum factor of safety of 2.5 will be used to restrict settlements at allowable loads to 1 inch.*

$Q_B = A_b\, q_{br}$

where

A_b = Base area, square feet

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Cohesionless Soils (Sand)

Side Resistance (Skin Friction)

- For skin friction, the Beta β method is used here.

$$f_{sz} = \beta\sigma_z \leq 4.0\ ksf$$

where

f_{sz} = Ultimate load transfer in skin friction at depth *Z*, *ksf*.

z = Depth below grade, feet.

σ_z = Vertical effective stress at depth *z*, *ksf* (soil density *x* depth)

β = $1.5 - 0.135\sqrt{z}$ $\quad 1.2 \leq \beta \geq 0.25$

$$Q_s = \int_0^L \beta\sigma_z\,dA$$

where

dA = Differential area of the perimeter, square feet.

L = Embedment depth below ground surface, feet.

For lateral deflections greater than 0.2 inches, the skin friction down to the point of zero deflection is neglected.

Base Resistance (End Bearing)

- For end bearing resistance, short term settlements are the governing criteria. Sand tends to loosen at the bottom of an excavation. Also there appears to be some densification beneath the base. Since a large amount of settlement cannot be tolerated, the limiting values for end bearing resistance is restricted to a downward base movement of 5 percent of base diameter.

Note: *Bells are not recommended in cohesionless soils. They are difficult to construct.*

- Ultimate end bearing stress, q_b, is a function of the uncorrected field values on standard penetration tests, *N*.

- For 5 percent base diameter settlement:

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$O \le N \le 75 \qquad q_b = 1.2N\ ksf.$

$N \le 75 \qquad q_b = 90.0\ ksf.$

q_b will be reduced when the base diameter, B_b, is greater than 50 inches.

$$\text{Reduced } q_{br} = \frac{50}{12B_b} q_b$$

$$Q_B = A_b\ q_{br}$$

where

A_b = Base area, square feet.

B_b = Bell diameter, feet

Note: *A minimum factor of safety of 2.5 will be used to restrict settlement to 1 inch.*

AXIAL TENSION LOAD

- For axial tension load or uplift, field test results are limited. A higher factor of safety will be used.
- The general equation for computing the ultimate tension capacity of drilled pier is as follows:

$$Q_{TT} = Q_s + Q_{BELL} + W_C$$

where

Q_{TT} = Ultimate tension capacity of drilled pier foundation.

Q_S = Ultimate capacity in side or skin resistance.

Q_{BELL} = Ultimate bell pullout resistance

W_C = Weight of concrete (shaft and bell) and soil above bell.

- To obtain service load allowable, a global factor of safety of 3 to 4 will be applied. Alternatively individual factors of safety could be used.

  – Safety factor skin resistance = 4.
  – Safety factor bell pullout = 3 to 5.
  – Safety factor weight of pier = 1.5.

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$$Q_{Allowable\,Uplift} = \frac{Q_s}{4} + \frac{Q_{BELL}}{3\,to\,5} + \frac{W_c}{1.5}$$

Straight Pier

- The ultimate capacity is computed from skin resistance and weight of pier. The computation of skin resistance is the same as for compressive loading.

Belled Pier Foundation

- One commonly used method is the assumption of a vertical fictitious cylinder of soil of bell diameter B_b above the bell and then computing skin (frictional) resistance along this surface. The computation of skin resistance is same as for compressive loading. By this method,

  Q_{TT} = $Q_{SB} + W_S + W_{con}$

  where

  Q_{SB} = Ultimate capacity in friction or skin resistance using bell diameter B_b along entire cylinder length.

  W_S = Weight of soil within the assumed fictitious cylinder.

  W_{con} = Weight of concrete.

- A minimum factor of 3 will be used with no increase in allowable for transient loads.

Note: *Axial resistance due to skin friction must neglect any skin friction for the depth of frost. Frost causes jacking forces on piers. The minimum depth to prevent this must be 2- 1/2 times the frost depth.*

LATERAL LOAD

General

- The analysis of laterally loaded piers is a complicated soil structure interaction problem. Closed form solutions to predict deflection and reaction are available. However, these solutions assume a linear elastic soil response which may result in erroneous results. In order to correctly model the soil response, a nonlinear load transfer method is required. One such method is the P-y curve method.

- Pier behavior (Attachment 04) under lateral load is governed by the flexural stiffness of the pier relative to the stiffness of the soil surrounding the upper portion of the pier. A stiff or a short pier, rotates as a rigid element around a fixed point below grade. Failure for a stiff or short pier occurs when the ultimate passive resistance of soil is exceeded. For a flexible or long pier, the flexural capacity of the pier is the limiting criteria. Failure for a long pier occurs when the bending moment in the pier is equal to the moment capacity of pier. The full passive resistance in the soil along the entire length of pier is not developed for long piers.

- The lateral capacity of a pier also depends on the fixity at the top of the pier.
- At working load levels, bells, or underreams have no influence on allowable lateral capacity. At ultimate soil resistance, bells have a marginal and insignificant effect on lateral resistance.
- Computation of pier lateral capacity is based on ultimate soil resistance with a factor of safety applied for allowable loads. The minimum factor of safety is 2. Also, a limiting factor in computation of lateral capacity is the pier deflection at grade. This should be limited to 0.25 inches at working loads. (Higher values will reduce axial skin friction capacity).

Broms Method (Hand Calculations)

- This is a simple but comprehensive method for computing the lateral capacity of piers by hand. This method uses the linear theory of subgrade reaction and gives an approximate solution that is good for substantial portion of drilled piers designed at Fluor. The method uses ultimate soil resistance and application of a factor of safety. The results are comparable to nonlinear computerized solution at loads of one-third to one-half of the ultimate soil resistance. At these loads, deflections at the ground line are small and the linear theory of subgrade reaction ($P = ky$) is valid.
- This method characterizes pier behavior as short or long and failure modes depends on pier head fixity according to Attachment 04.

P-y Curve Method (Nonlinear Computerized Calculations)

- The P-y curve is a mathematical representation of the soil reaction versus pier deflection.
- Soil reaction is a function of pier deflection and pier deflection depends on soil reaction. Therefore, the solution involves equations of equilibrium and compatibility that need to be satisfied. This solution is difficult to solve by hand and requires use of computer programs.
- Attachment 13 shows a model of laterally loaded deep pier. Soil is replaced by a series of mechanisms that models soil response. Soil response, P, is a function of lateral deflection, y and depth, x.

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- P-y curves for soil will be obtained from the Geotechnical Report. From experimental results and theory, P-y curves, both for static and cyclic loading are available for the following situations:

  – Soft Clay Below Water (Matlock, 1970).
  – Stiff Clay Below Water (Reese, 1973).
  – Stiff Clay above Water Table (Welch and Reese, 1972).
  – Sand (Cox, 1974).
  – Unified Criteria for Clay (Sullivan, 1979).

- The Geotechnical Report should clarify if the above P-y curves are applicable.

- The P-y method will be used for structures carrying significant loads. The Lead Structural Engineer will decide the structures that require detailed lateral load analysis. Close collaboration with the Geotechnical Consultant is very essential. Computer programs allow investigation of the influence of a large number of parameters such as loading, geometry, pier penetration, soil properties, pier/superstructure interaction, and buckling with minimum difficulty.

LATERAL LOAD DESIGN USING BROMS METHOD

Limitations

- This method assumes a uniform soil profile.

- This method uses the linear theory of subgrade reaction and is applicable to isolated piers such as 6 to 8 pier diameter spacing.

- This method will be used on existing sites where previous Geotechnical Reports have not adequately addressed allowable lateral load capacity.

- This method will be used with close collaboration with the Geotechnical Consultant. The required parameters will be obtained from the Geotechnical Report.

- This method does not give shear, moments, and deflections along the pier length at allowable loads. The method uses the concept of ultimate soil resistance at failure and an application of factor of safety to obtain allowable lateral load. Concrete design is on the conservative side. For additional discussion, refer to sample computations.

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Cohesive Soils (Clays)

General

- Broms assumes soil and pier as elastic materials and the formulation is based on results from saturated cohesive clays that are over consolidated. The method assumes ultimate soil resistance as 9 times the undrained soil shear strength (C_u) times the pier diameter (D) such as 9 $C_u D$, regardless of depth. Soil resistance at top 1.5 pier diameters is neglected. (Refer to Attachment 05).

- The ultimate shear strength of fissured clays is less than 9 C_u; hence, this procedure may lead to slightly unconservative answers for such soils.

  The behavior of pier depends on the dimensionless length βL:

$$\beta = \left(\frac{kD}{4E_p I_p} \right)^{0.25}$$

k = Modulus of subgrade reaction for cohesive soil, lbs/cubic inch.
D = Pier diameter, inch.
E_p = Pier modulus of elasticity, lbs/square inch.
I_p = Pier moments of inertia, inch^4.
L = Pier embedment length, inch.

- Terzaghi's recommendations for k are as follows:

Unconfined Compressive Strength	k	
(Tons/Square Feet)	(Tons/Cubic Feet)	(Lbs/Cubic Inch)
1-2	75	87
2-4	150	173
>4	300	347

- For Free Headed Piers:

  – Short If βL < 1.5 (Refer to Attachment 05, Figure 2)
  – Long If βL > 2.5 (Refer to Attachment 05, Figure 3)
  – Transition between βL 1.5 to 2.5.

- For Fixed Headed Piers:

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- Short — If $\beta L < 0.5$ (Refer to Attachment 05, Figure 4)
- Intermediate — $0.5 < \beta L < 1.5$ (Refer to Attachment 05, Figure 5)
- Long — If $\beta L > 1.5$ (Refer to Attachment 05, Figure 6)

- The capacity of short pier depends on soil shear strength, whereas capacity of long pier depends on ultimate concrete capacity of pier section.

- Pier behavior is governed by soil properties at the top. Average values at the top will be used.

Free Headed

- Refer to Attachments 5, Figure 2 (short) and Figure 3 (long).

- Short Pier

  For a short pier (Refer to Attachment 05, Figure 2) failure takes place when the pier rotates and ultimate soil resistance has developed along entire pier length.

  Point of maximum moment and zero shear is at distance ($1.5D + f$) below grade.

  From statics:

$$f = \frac{P_{ULT}}{9C_u D}$$

$$M_{maxp} = P_{ULT}\,(e + 1.5D + 0.5f)$$

$$M_{maxp} = 2.25\ C_u D g^2$$

$$L = 1.5D + f + g$$

  Substituting will produce a quadratic equation in P_{ULT} which can be solved. Alternatively, Attachment 06 could be used. A factor of safety will be applied to obtain allowable P_a.

  The lateral deflection at ground line is as follows:

$$Y_O = \frac{4P_a\left(1 + 1.5\frac{e}{L}\right)}{kDL}$$

  Alternatively, Attachment 08 could be used.

- Long Pier

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For a long pier (Refer to Attachment 05, Figure 3), failure occurs with the formation of a plastic hinge in the pier (material failure) at a depth of (1.5*D* + *f*) below ground surface.

From statics:

$$f = \frac{P_{ULT}}{9C_u D}$$

$$P_{ULT} = \frac{M_{\max}}{(e + 1.5D + 0.5f)}$$

where

M_{max} = Moment capacity of pier section, unfactored.

Substituting will produce a quadratic equation in P_{ULT} which can be solved. Alternatively, Attachment 07 could be used. A factor of safety will be applied to obtain allowable P_a.

The lateral deflection at ground line is as follows:

$$Y_o = \frac{2P_a\beta(e\beta + 1)}{kD}$$

Alternatively, Attachment 08 could be used.

Note: *For 1.5 < βL < 2.5, compute P_{ULT} as if it were a short pier. Then compute maximum moment, M_{maxp}. If this exceeds pier moment capacity, then use long pier equations.*

Fixed Headed

- Refer to Attachment 05, Figure 4 (Short), Figure 5 (Intermediate), and Figure 6 (Long).
- Short Pier

For a short pier (Attachment 05, Figure 4), failure takes place when the pier moves horizontally and ultimate soil resistance is developed along entire pier length. (Except top 1.5 pier diameters).

From statics:

P_{ULT} = $9\, C_u D\, (L - 1.5D)$

The maximum moment occurs at the top and will be less than moment capacity of

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pier section.

$$M_{maxn} = P_{ULT}\,(0.5L + 0.7D) \le M_{max}$$

where

M_{max} = Moment capacity of pier section, unfactored.

A factor of safety will be applied to obtain allowable P_a.

The lateral deflection at ground line is as follows:

$$Y_O = \frac{P_a}{kDL}$$

Alternatively, Attachment 08 could be used.

- Intermediate Pier

For an intermediate pier (Refer to Attachment 05, Figure 5), failure takes place with the formation of a plastic hinge at the top (material failure) allowing the pier to rotate.

From statics:

$$f = \frac{P_{ULT}}{9C_u D}$$

$$M_{maxp} = P_{ULT}\,(1.5D + 0.5f) - M_{max}$$

$$M_{maxp} = 2.25\,C_u\,Dg^2$$

$$L = 1.5D + f + g$$

Substituting will produce a quadratic equation in P_{ULT} which can be solved. A factor of safety will be applied to obtain allowable P_a.

The lateral deflection at ground line, can be obtained from Attachment 08.

- Long Pier

For a long pier (Refer to Attachment 05, Figure 6), failure occurs with the formation of a plastic hinge at the maximum negative moment (top of pier) and another hinge at the maximum positive moment located at a distance (1.5D + f) below grade. This mode of failure is material failure of pier section.

From statics:

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$$f = \frac{P_{ULT}}{9C_u D}$$

$$P_{ULT} = \frac{2M_{\max}}{1.5D + 0.5f}$$

where

M_{max} = Moment capacity of pier section, unfactored.

Substituting will produce a quadratic equation in P_{ULT} which can be solved. Alternatively, Attachment 07 could be used. A factor of safety will be applied to obtain allowable P_a.

The lateral deflection at ground line is as follows:

$$Y_O = \frac{P\alpha\beta}{kD}$$

Alternatively, Attachment 08 could be used.

Cohesionless Soils (Sands)

General

- The method is based on the assumption that the lateral modulus of subgrade reaction increases linearly with depth and decreases linearly with width ($k_h = n_h z/D$). Also the ultimate soil lateral resistance is equal to 3 times the passive rankine earth pressure.
- The assumed distribution of lateral earth pressure at failure is shown on Attachment 09.
- The ultimate soil reaction, Q, per unit length of pier is:

$$Q = 3D\,\gamma z\,K_p$$

where

D = Diameter of pier.

γ = Unit weight of soil.

z = Depth below ground surface.

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K_p = Rankine coefficient of passive earth pressure = $\frac{1+\sin\varphi}{1-\sin\phi}$

ϕ = Soil angle of internal friction.

- Behavior of pier depends on the dimensionless length λL:

$$\lambda = \left(\frac{n_h}{E_p I_p}\right)^{0.2}$$

n_h = Coefficient of lateral subgrade reaction for a long pier with a width of unity at a depth of unity, lbs/cubic inch.
I_p = Pier moment of inertia, inch[4].
E_p = Pier modulus of elasticity, lbs/square inch.
k_h = Coefficient of horizontal subgrade reaction, lbs/cubic inch.
L = Pier embedment length, inch.

Terzaghi's recommendation for n_h are:

Sand Relative Density	n_h (Tons/cubic feet) Above Ground Water Table	n_h (Tons/cubic feet) Below Ground Water Table
Loose, 4 < N < 10	7	4
Medium, 10 < N < 30	21	14
Dense, N > 30	56	34

- Piers are classified as:

  – Short If $\lambda L < 2.0$ Attachment 09, Figure 2, Figure 4
  – Long If $\lambda L > 4.0$ Attachment 09, Figure 5, Figure 6

  Between λL 2.0 and 4.0, pier behavior is transition for Free Headed and Intermediate for Fixed Headed. (Refer to Attachment 09, Figure 5).

- The capacity of short pier depends on soil properties whereas capacity of long pier depends on ultimate concrete capacity of pier section.

- Pier behavior is governed by soil properties near the top. Average properties at 3 to 5 pier diameters from top will be used.

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DRILLED PIER FOUNDATIONS

Free Headed

- Refer to Attachment 09, Figure 2 (Short) and Figure 3 (Long).
- Short Pier

  For a short pier (Attachment 09, Figure 2), failure takes place when the pier rotates and ultimate soil resistance has developed along entire pier length.

  Point of maximum moment and zero shear is at a distance *f* below grade.

  From statics:

$$P_{ULT} = \frac{\gamma D L^3 K_p}{2(e+L)}$$

$$f = 0.816 \left(\frac{P_{ULT}}{\gamma D K_p}\right)^{1/2}$$

$$M_{maxp} = P_{ULT}\ (e + f) - P_{ULT}$$

$$L = f + g$$

  The above equations can be used for computing P_{ULT} and a factor of safety will be applied to obtain allowable P_a. Alternatively, Attachment 10 could be used.

  The lateral deflection at ground line:

$$Y_O = \frac{18 P_a \left(1 + 1.33\frac{e}{L}\right)}{L^2 n_h}$$

  Alternatively, Attachment 12 could be used.

- Long Pier

  For a long pier (Attachment 09, Figure 3), failure occurs with the formation of a plastic hinge in the pier (material failure) at a depth f below grade.

  From statics:

$$f = 0.816 \left(\frac{P_{ULT}}{\gamma D K_p}\right)^{1/2}$$

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DRILLED PIER FOUNDATIONS

$$P_{ULT} = \frac{M_{\max}}{e + 0.544\left(\frac{p}{\gamma D K_p}\right)^{1/2}}$$

where

M_{max} = Moment capacity of pier section unfactored.

Substituting will produce a quadratic equation in P_{ULT} which can be solved. Alternatively Attachment 11 could be used. A factor of safety will be applied to obtain allowable P_a.

The lateral deflection at ground line:

$$Y_O = \frac{2.40 P_a}{n_h^{3/5} \left(E_p I_p\right)^{2/5}}$$

Alternatively, Attachment 12 could be used.

Note: *For 2.0 < λL < 4.0, compute P_{ULT} and maximum moment, M_{maxp} as if it were a short pier. If the maximum moment exceeds moment capacity of pier, then use long pier equations.*

Fixed Headed

- Refer to Attachment 09, Figure 4 (Short), Figure 5 (Intermediate), and Figure 6 (Long).

- Short Pier

  For a short pier (Refer to Attachment 09, Figure 4), failure takes place when the pier moves horizontally and ultimate soil resistance is developed along entire pier length.

  From statics:

  P_{ULT} = $1.5\, \lambda\, L^2 D\, K_p$

  The maximum moment occurs at the top and will be less than moment capacity of the pier section.

  M_{maxn} = $0.67\, P_{ULT}\, L \leq M_{max}$

  where

  M_{max}= Moment capacity of pier section, unfactored.

A factor of safety will be applied to obtain allowable P_a.

The lateral deflection at ground line:

$$Y_O = \frac{2P_a}{L^2 n_h}$$

Alternatively, Attachment 12 could be used.

- Intermediate Pier

For an intermediate pier (Attachment 09, Figure 5), failure takes place with the formation of a plastic hinge at the top (material failure) allowing the pier to rotate.

From statics:

$$P_{ULT} = 0.5\,\lambda\, D\, L^2\, K_p + \frac{M_{\max}}{L}$$

where

M_{max}= Moment capacity of pier section, unfactored.

The above equation is valid only if the maximum positive moment at distance f below grade is less than M_{max}.

$$M_{maxp} = P_{ULT}\,(e + 0.67f) - M_{max}$$

$$f = 0.816\left(\frac{P_{ULT}}{\gamma D K_p}\right)^{1/2}$$

The above equation can be used for computing P_{ULT} and a factor of safety will be applied to obtain allowable P_a.

The lateral deflection at ground, Y_O can be obtained from Attachment 12.

- Long Pier

For a long pier (Attachment 09, Figure 6), failure occurs with the formation of a plastic hinge at the maximum negative moment (top of pier) and another hinge at the maximum positive moment located at a distance f below grade. This mode of failure is material failure of pier section.

From statics:

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DRILLED PIER FOUNDATIONS

$$P_{ULT} = \frac{2M_{\max}}{e + 0.544\left(\frac{p}{\gamma D K_p}\right)^{1/2}}$$

where

M_{max} = Moment capacity of pier section, unfactored.

For $e = o$ such as P_{ULT} at bottom of pier cap:

$$P_{ULT} = \left[3.676 M_{\max} \left(\gamma D K_p\right)^{1/2}\right]^{2/3}$$

The point of maximum positive moment, f:

$$f = 0.816 \left(\frac{P_{ULT}}{\gamma D K_p}\right)^{1/2}$$

The previous equation can be used to compute P_{ULT} and a factor of safety will be applied to obtain allowable P_a. Alternatively, Attachment 11 could be used to compute P_{ULT}.

The lateral deflection at ground line is as follows:

$$Y_O = 0.93 \frac{P_a}{n_h^{3/5}\left(E_p I_p\right)^{2/5}}$$

Alternatively, Attachment 12 could be used.

GROUP EFFECTS

- The design methods outlined in previous sections are for single isolated piers at a spacing of 6 to 8 pier diameters. Group effects significantly reduce allowable capacities and are an important consideration.
- The minimum recommended pier spacing will be 3 pier diameters. This is also the minimum for constructibility of pier without significant problems.

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DRILLED PIER FOUNDATIONS

Axial

Cohesive

- For cohesive soils, the capacity of individual pier will be reduced to 0.7 at a spacing of 3 pier diameter. For greater spacing, interpolate from 1.0 at 8 pier diameter to 0.7 at 3 pier diameter spacing.
- Group effects also depend on the group configuration and settlements of the group. For example, the capacity of a group of 4 by 4 piers is less than a group of 2 by 2 piers. A simple block method could be used to compute group efficiency. The Geotechnical Report will provide recommendations.
- Where pier caps are used, the cap will be in firm contact with the soil or else the capacity will be further reduced to 0.67 the capacity at 3 pier diameter spacing.

Cohesionless

- For cohesionless soils, the capacity of individual piers is the same at a spacing of 3 pier diameters as an isolated pier.
- Where pier caps are used, the capacity of individual piers will be reduced to 0.67 the capacity at 3 pier diameter spacing regardless of whether the pier cap is in firm contact with the soil.
- For lesser spacing, block method could be used. The Geotechnical Report will provide recommendations.

Lateral

- Group effect of laterally loaded pier is a complex problem and is a function of geometrical layout, pier diameter, and spacing. Using the P-y method, the P-y curves could be modified by reducing *P* by a group efficiency factor. For the Broms method, the lateral capacity will be reduced by a group efficiency factor.
- The Geotechnical Report will provide recommendation on group efficiency factors. These factors are normally in the range of 0.62 for a 2 by 2 group at 3 pier diameter spacing to 0.44 for a 4 by 4 group.
- For high seismic zone, combination of cyclic loading and group action could very significantly reduce allowable lateral capacity. The P-y method computer programs are recommended for computing the lateral capacity for this case. This will be done with close collaboration with the Geotechnical Consultant.

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DRILLED PIER FOUNDATIONS

SETTLEMENTS

- One of the advantages of drilled piers is that they can carry larger loads with less settlement than for instance, spread footings. However, computation of a compression load depends on acceptable settlements. This is more pronounced when more than 50 percent of the load is resisted by the base. Long drilled piers resisting a good percentage of applied loads in skin resistance tend to settle the least and are more acceptable for performance of the structure.

- The settlement at the top of the pier is due to the elastic shortening of the shaft and the settlement at the base is due to base resistance and skin resistance.

- Short term settlement is of significant importance to prevent differential settlements. Simplified methods are available and may be required for long heavily loaded piers. Computer programs normally provide short term settlements.

- Long term settlement is complex to compute and is done by Geotechnical Consultants using consolidation and elastic half space theories.

- Both short and long term settlements are of significant importance for a successful design of drilled pier and close collaboration with Geotechnical Consultant is required.

PIER BUCKLING

- Potential pier buckling under high axial loads in weak soil will be investigated.

- Normally, soils having a standard penetration value of *N* equal or greater than 2 may be considered to provide adequate lateral support to prevent buckling.

- For additional details refer to: U.S. Naval Manual NAVFAC DM 7.2, May 1982.

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Structural Engineering
Guideline 000.215.1232: Driven Pile Foundations

Structural Engineering
Specification 000.215.02380: Drilled Caissons

ATTACHMENTS

Attachment 01:
Typical Drilled Pier Foundation

Attachment 02:
Typical Underream Shape

Attachment 03:
Design Procedure For Drilled Pier Foundation - Geotechnical

DRILLED PIER FOUNDATIONS

Attachment 04:
Failure Modes For Piles

Attachment 05:
Deflection, Soil Reaction, And Bending Moment Diagram For Piles In Cohesive Soil

Attachment 06:
Cohesive Soils - Ultimate Lateral Resistance Of Short Piles

Attachment 07:
Cohesive Soils - Ultimate Lateral Resistance Of Long Piles

Attachment 08:
Cohesive Soil - Lateral Deflection At Ground Surface

Attachment 09:
Deflection, Soil Reaction, And Bending Moment Diagram For Piles In Cohesionless Soils

Attachment 10:
Cohesionless Soil - Ultimate Lateral Resistance Of Short Piles

Attachment 11:
Cohesionless Soil - Ultimate Lateral Resistance Of Long Piles

Attachment 12:
Cohesionless Soil - Lateral Deflections At Ground Surface

Attachment 13:
Model Of A Deep Pile Foundation Under Lateral Loading Showing Concept Of Soil Response Curves

Attachment 14:
Explanation Of Portions Of Drilled Pier Not Considered In Computing Side Resistance

Attachment 15:
Crane Mounted Drilling Unit

Attachment 16:
Sample Design 1: Straight Pile In Cohesive Soil

Note: *Calculations for this attachment have not been updated to meet the requirements of the latest references and design codes.*

Attachment 17:
Sample Design 2: Belled Pier In Mixed Soil Profile

Note: *Calculations for this attachment have not been updated to meet the requirements of the latest references and design codes.*

Attachment 18:
Sample Design 3: Short Straight Pile In Sand

Note: *Calculations for this attachment have not been updated to meet the requirements of the latest references and design codes.*

Attachment 19:
Sample Design 4: Short Belled Pier In Clay

Note: *Calculations for this attachment have not been updated to meet the requirements of the latest references and design codes.*

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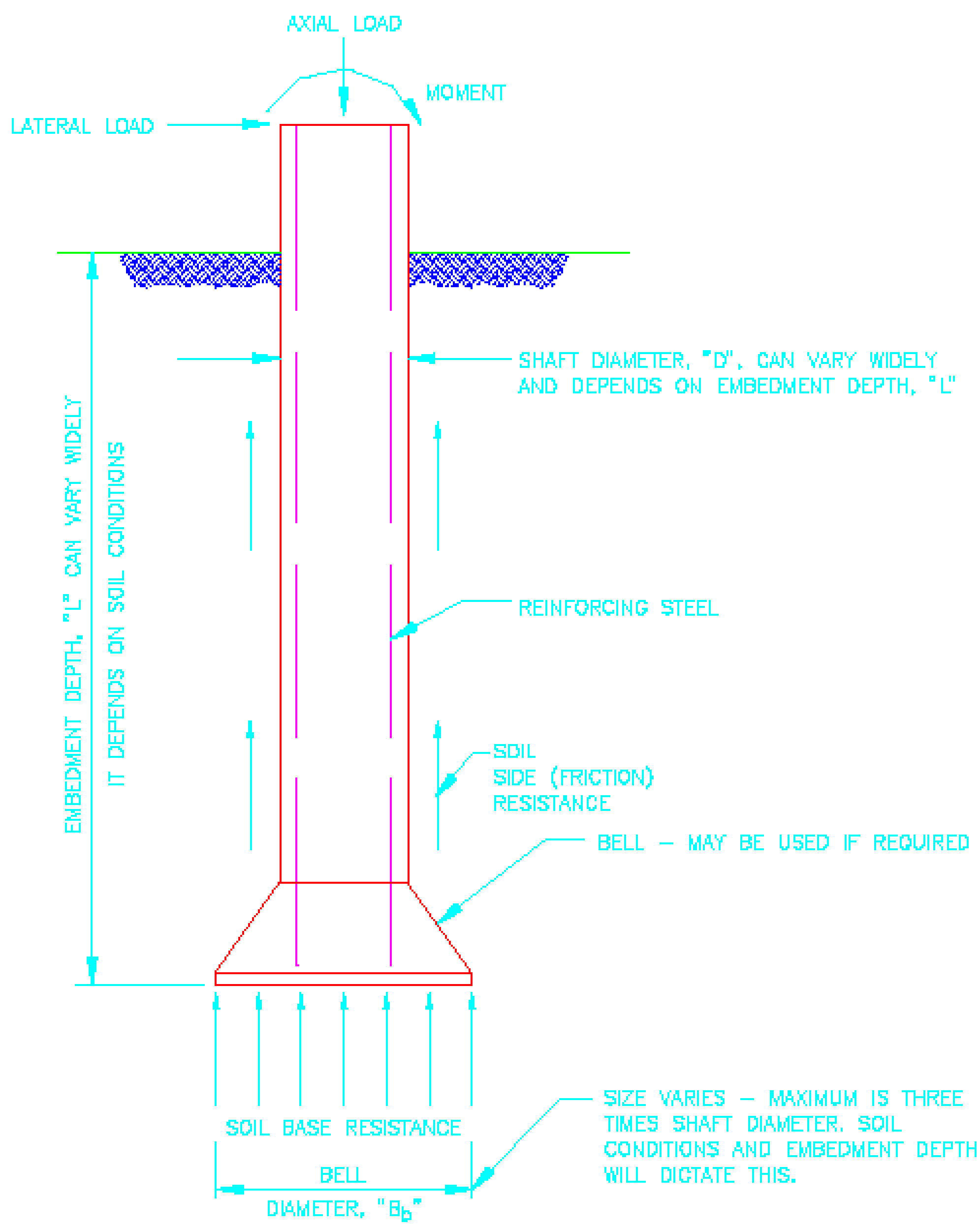
TYPICAL DRILLED PIER FOUNDATIONS



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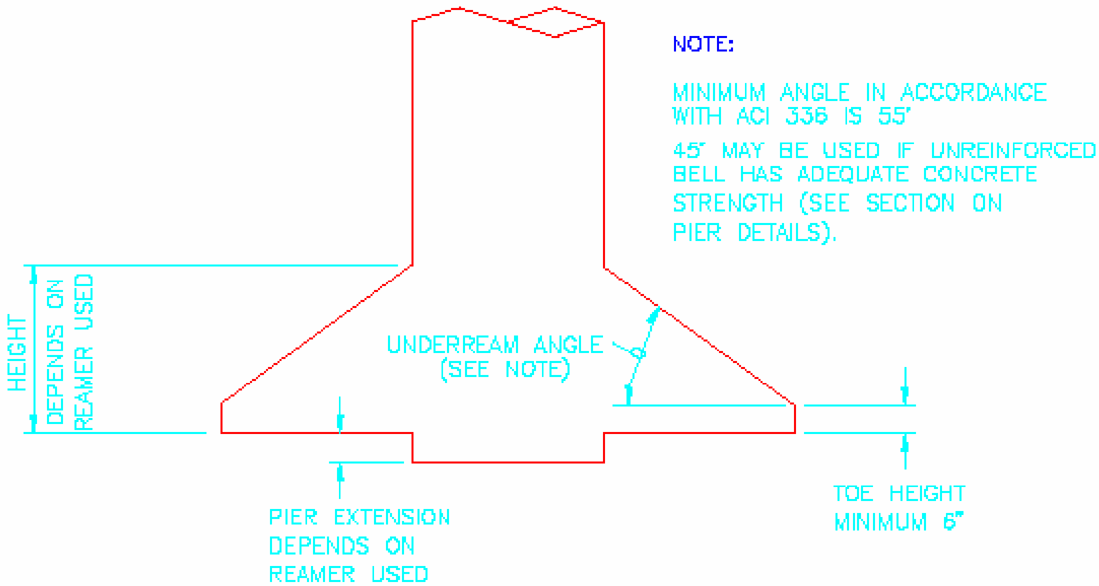
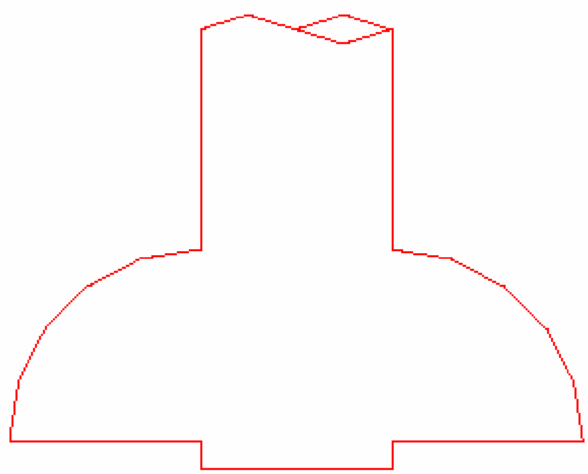
TYPICAL UNDERREAM SHAPE



UNDERREAM CUT WITH STANDARD REAMER



UNDERREAM CUT WITH BUCKET REAMER

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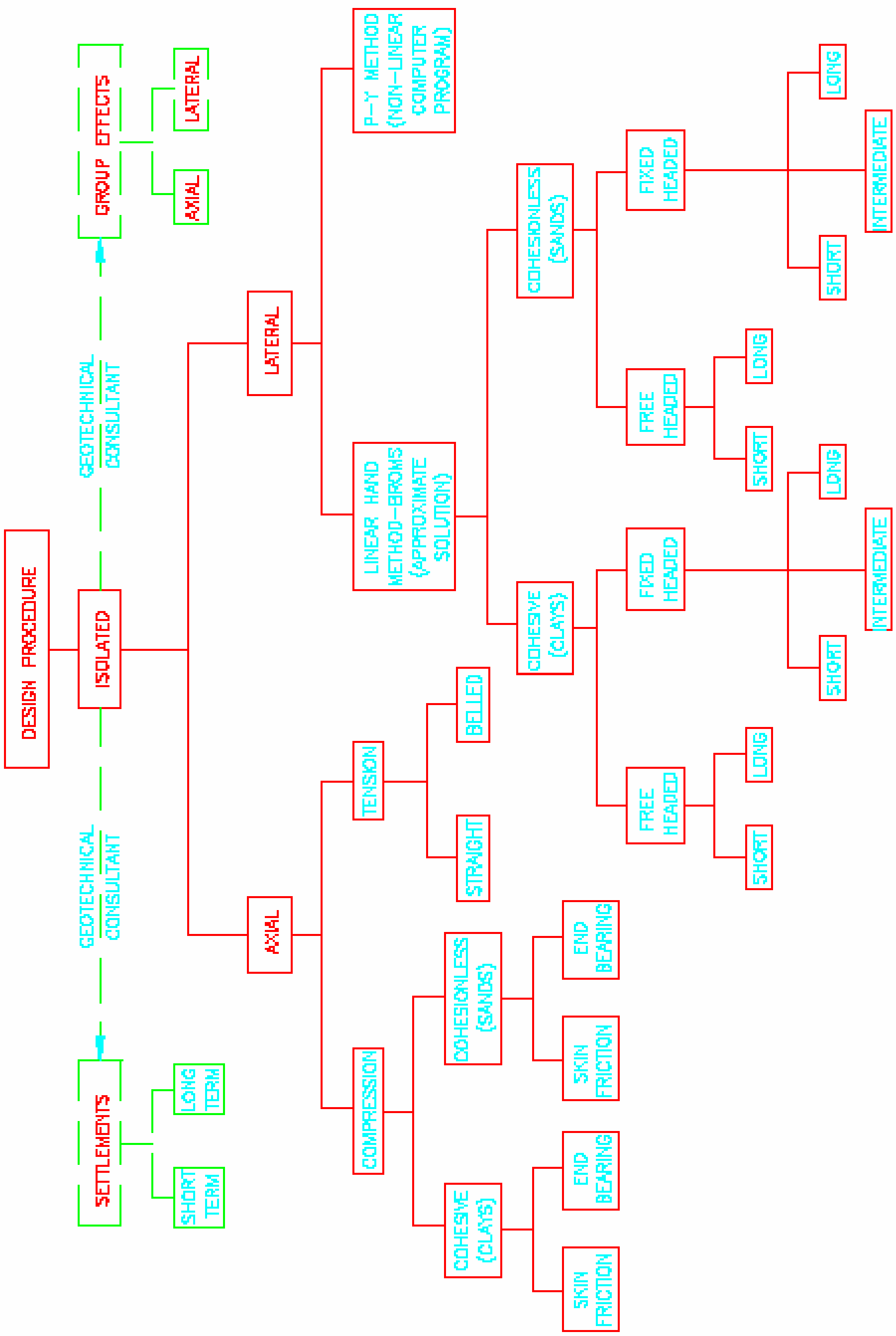
DESIGN PROCEDURE FOR DRILLED PIER FOUNDATION - GEOTECHNICAL



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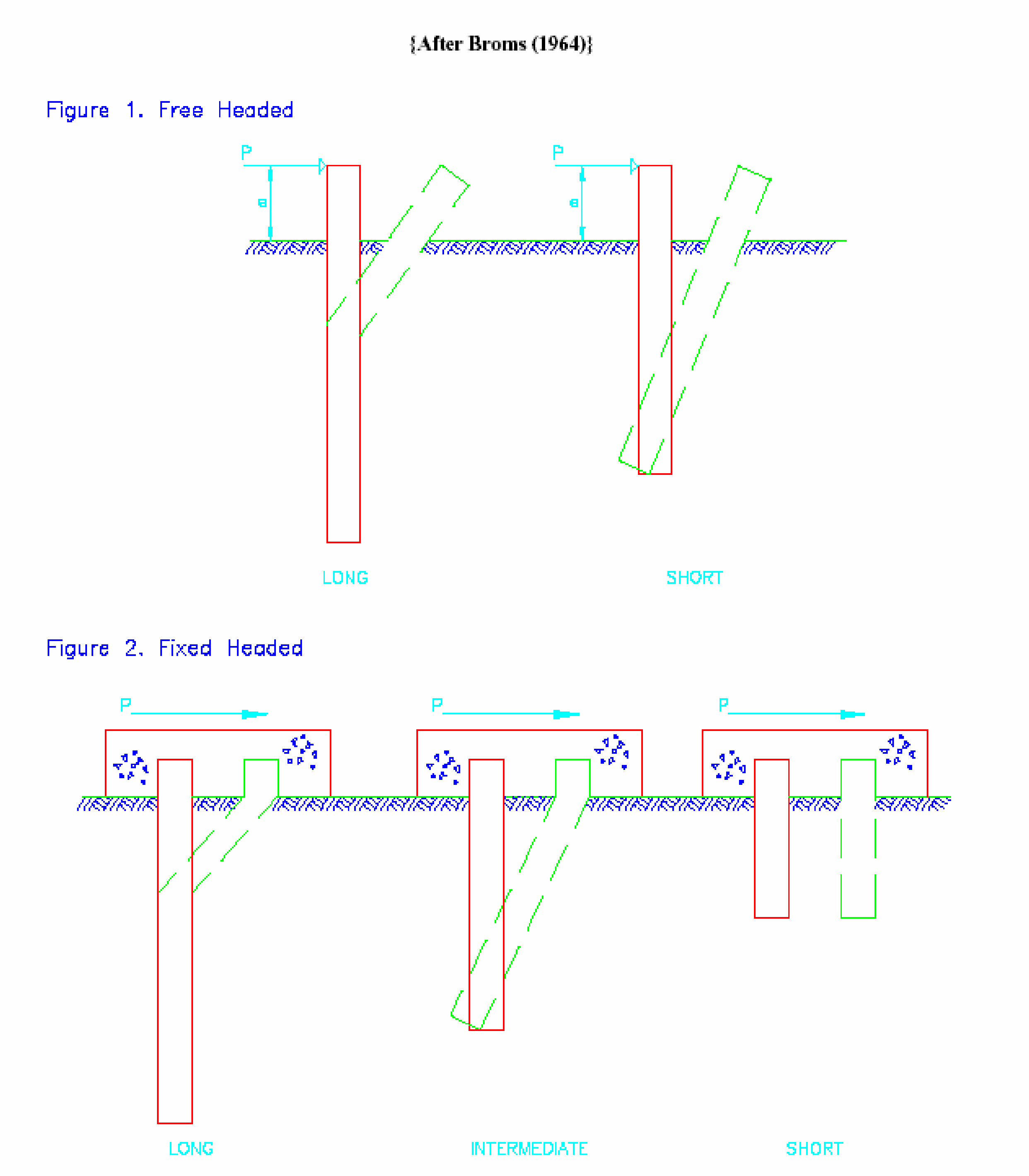
FAILURE MODES FOR PILES



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DEFLECTION, SOIL REACTION AND BENDING MOMENT DIAGRAM FOR PILES IN COHESIVE SOILS

{After Broms (1964)}

Figure 1: Distribution of Lateral Earth Pressures in Cohesive Soil

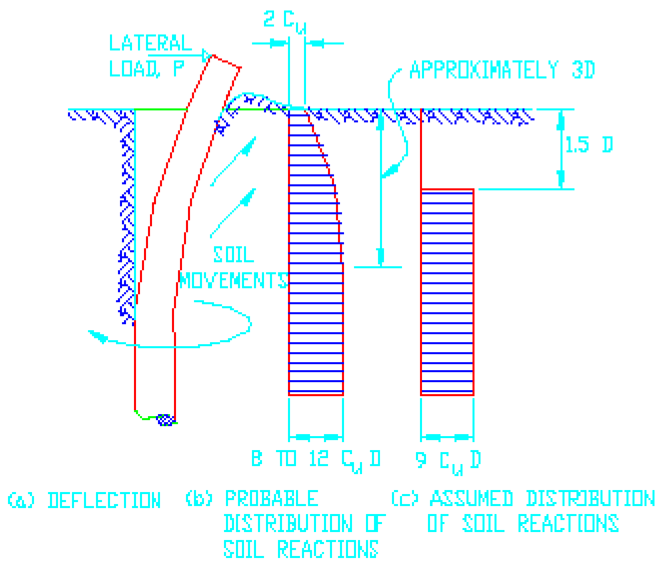
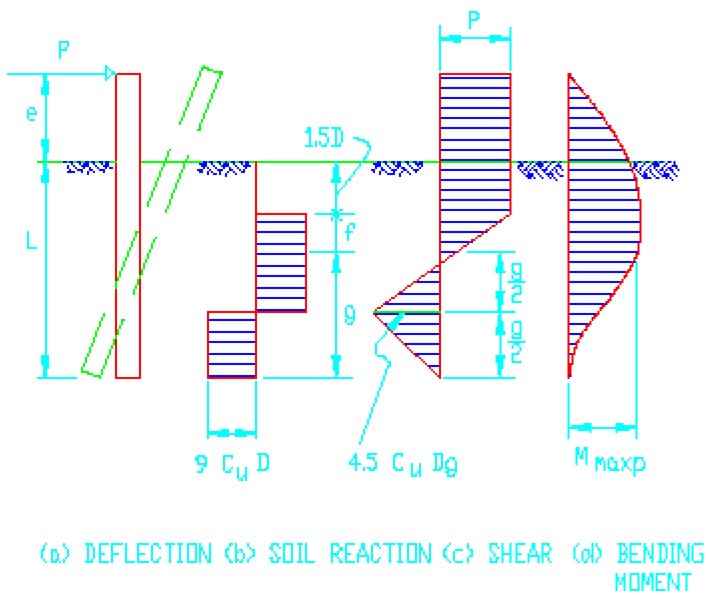


Figure 2: Short Free-Headed Pile



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DEFLECTION, SOIL REACTION AND BENDING MOMENT DIAGRAM FOR PILES IN COHESIVE SOILS

Figure 3: Long Free-Headed Pile

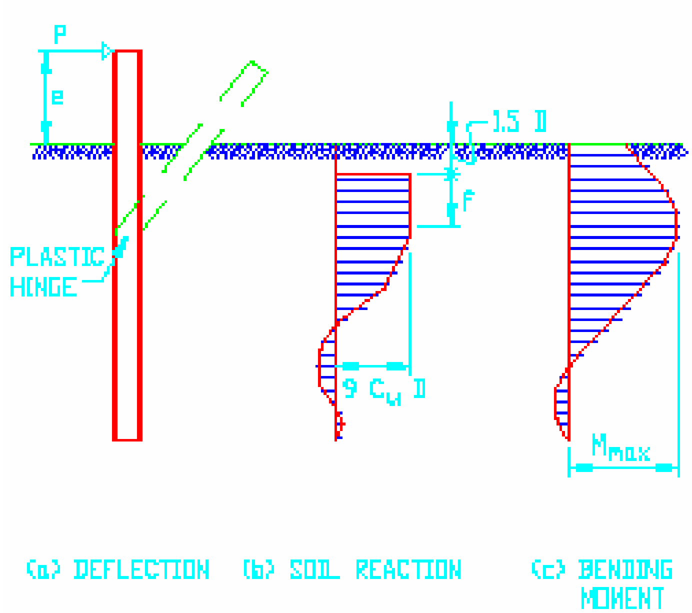
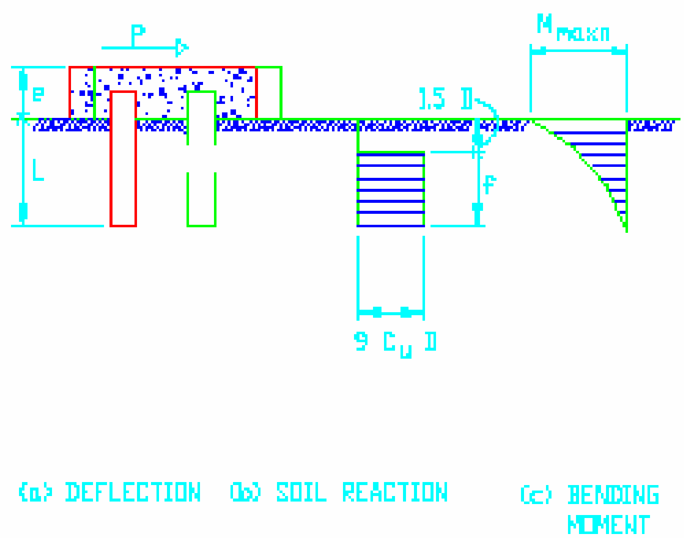


Figure 4: Short Fixed-Headed Pile



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DEFLECTION, SOIL REACTION AND BENDING MOMENT DIAGRAM FOR PILES IN COHESIVE SOILS

Figure 5: Intermediate-Length Fixed-Headed Pile

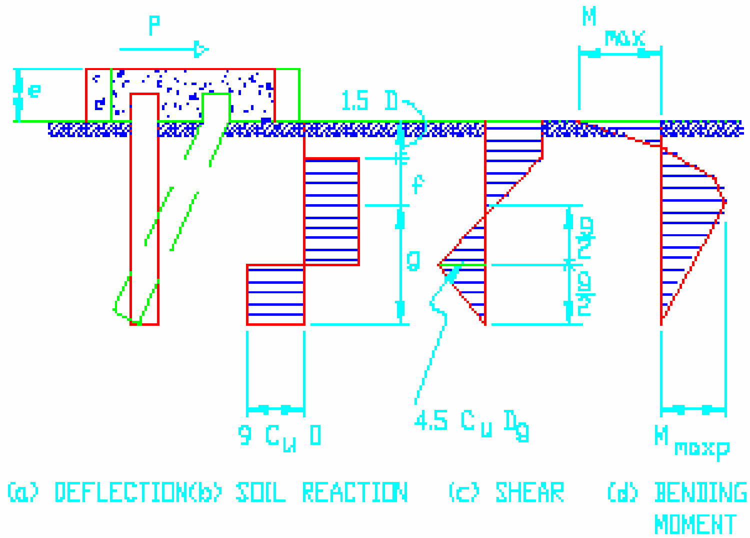
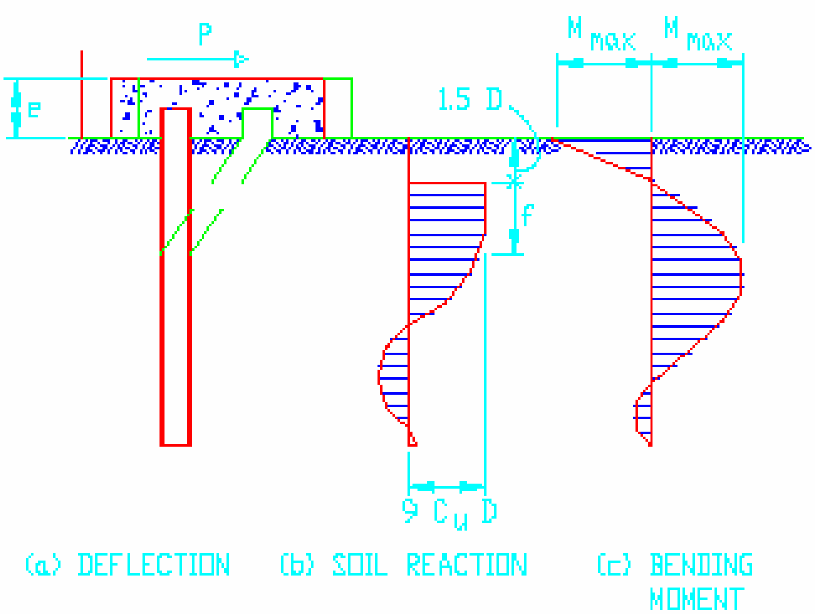


Figure 6: Long Fixed-Headed Pile



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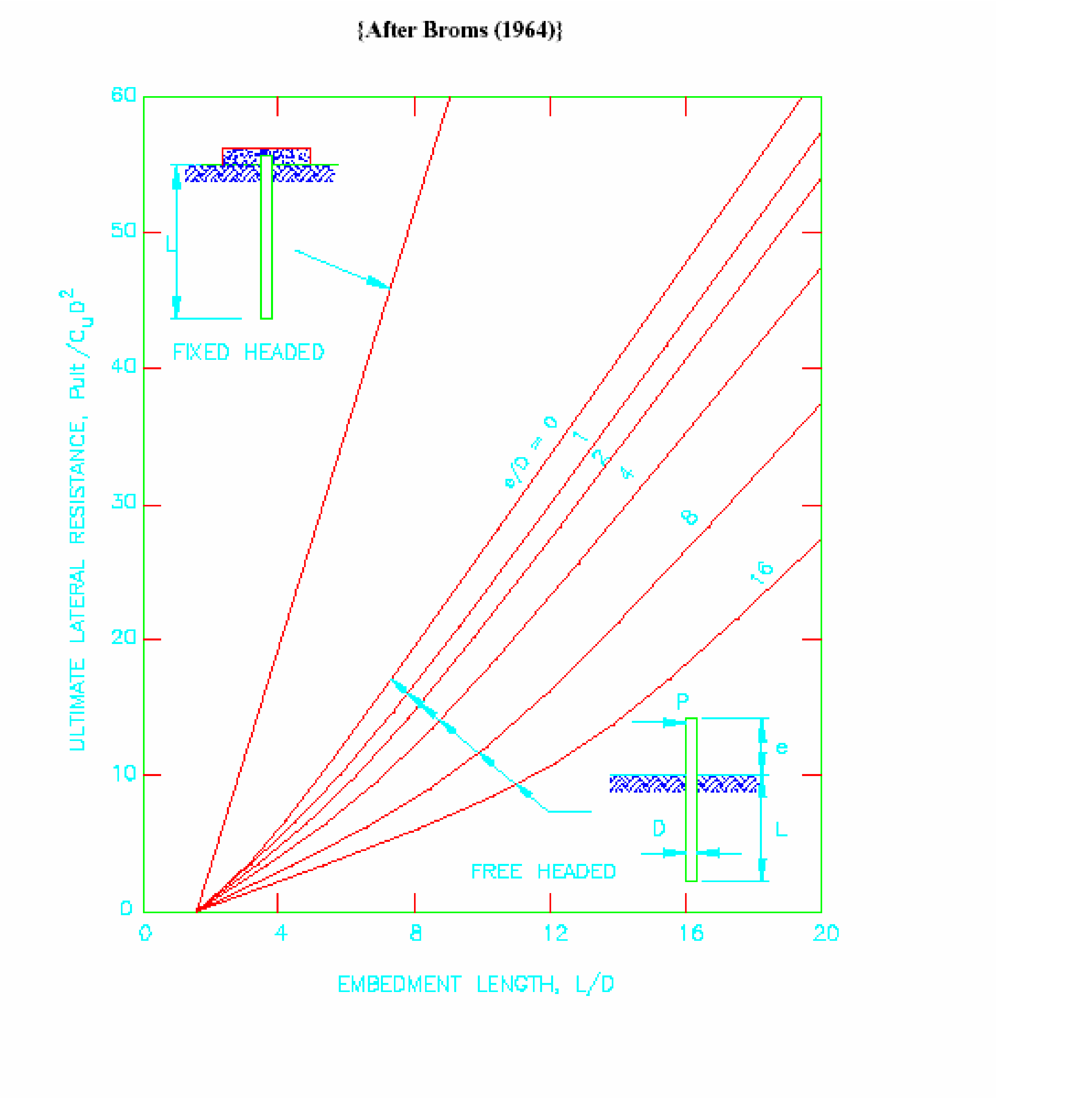
COHESIVE SOILS - ULTIMATE LATERAL RSISTANCE OF SHORT PILES



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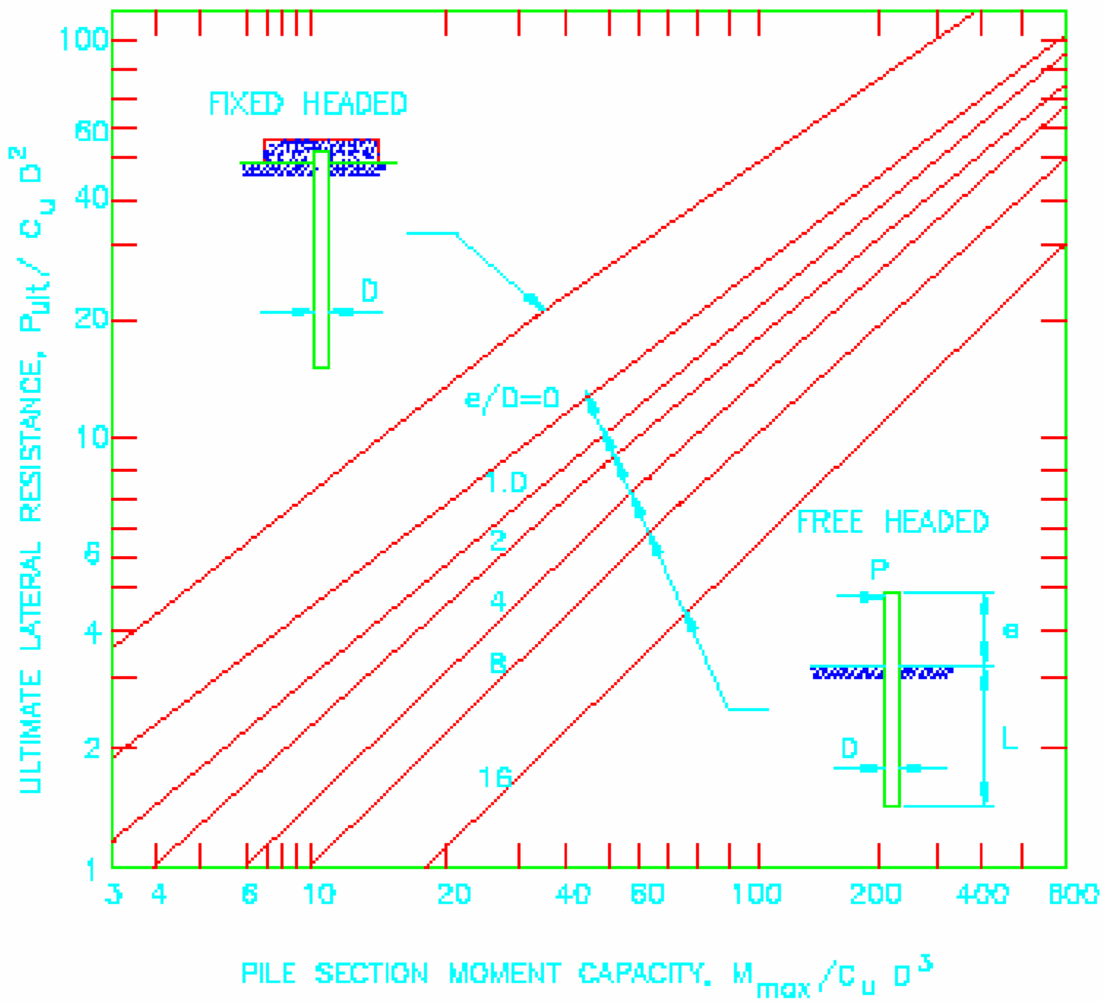
COHESIVE SOIL - LATERAL DEFLECTION AT GROUND SURFACE

{After Broms (1964)}



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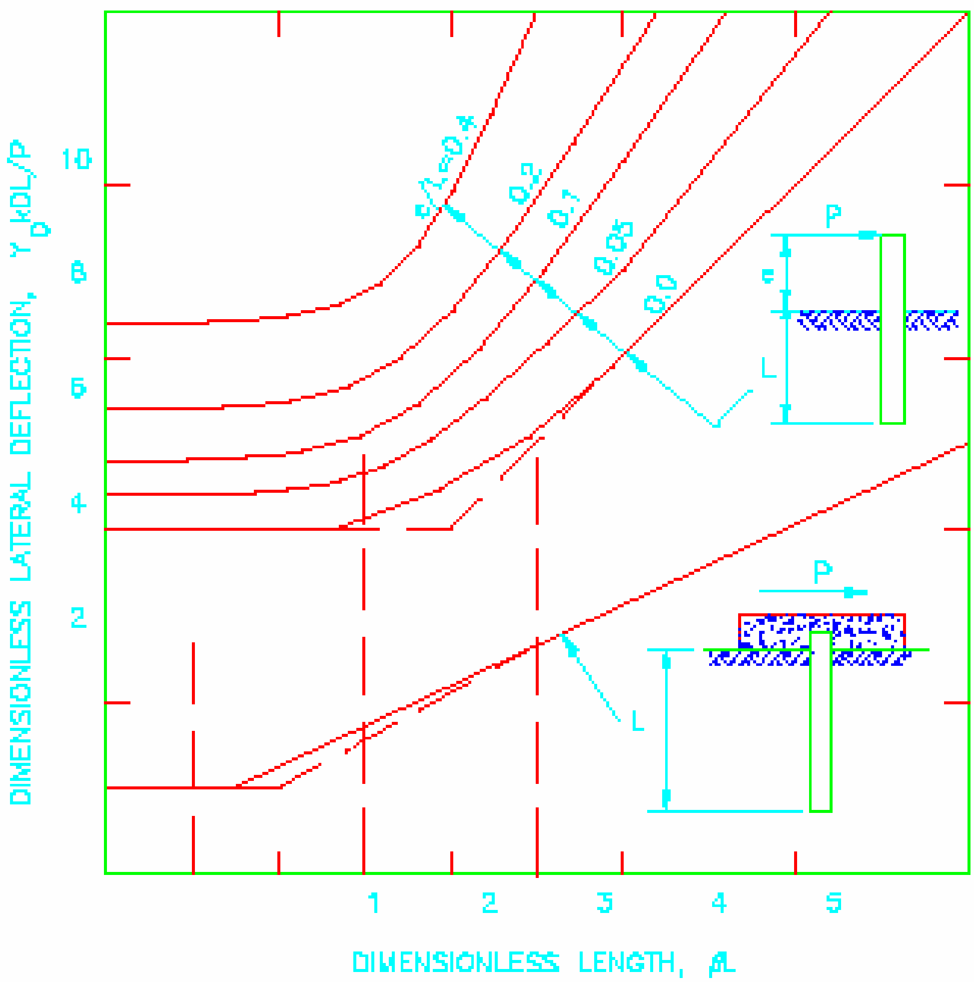
COHESIVE SOILS - ULTIMATE LATERAL RESISTANCE OF LONG PILES

{After Broms (1964)}



FLUOR

DEFLECTION, SOIL REACTION AND BENDING MOMENT DIAGRAM FOR PILES IN COHESIONLESS SOILS

{After Broms(1964)}

Figure 1: Assumed Distribution of Soil Reactions in Cohesionless Soil

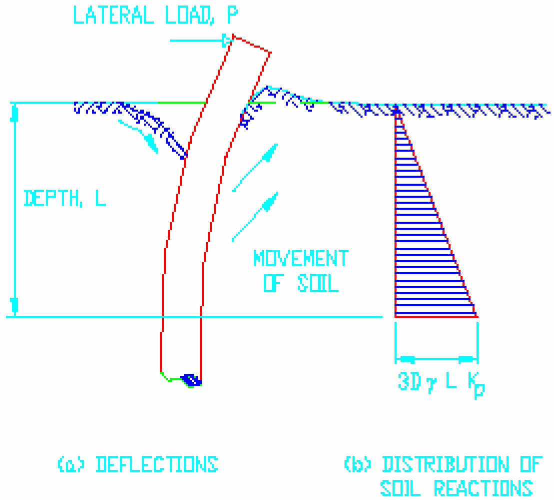
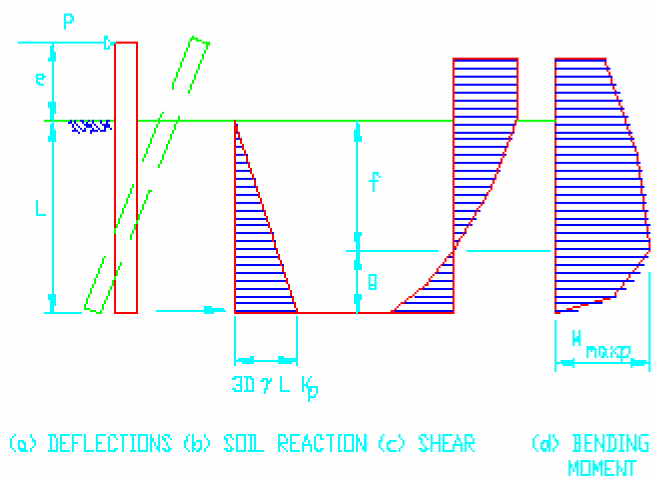


Figure 2: Short Free-Headed Pile



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DEFLECTION, SOIL REACTION AND BENDING MOMENT DIAGRAM FOR PILES IN COHESIONLESS SOILS

Figure 3: Long Free-Headed Pile

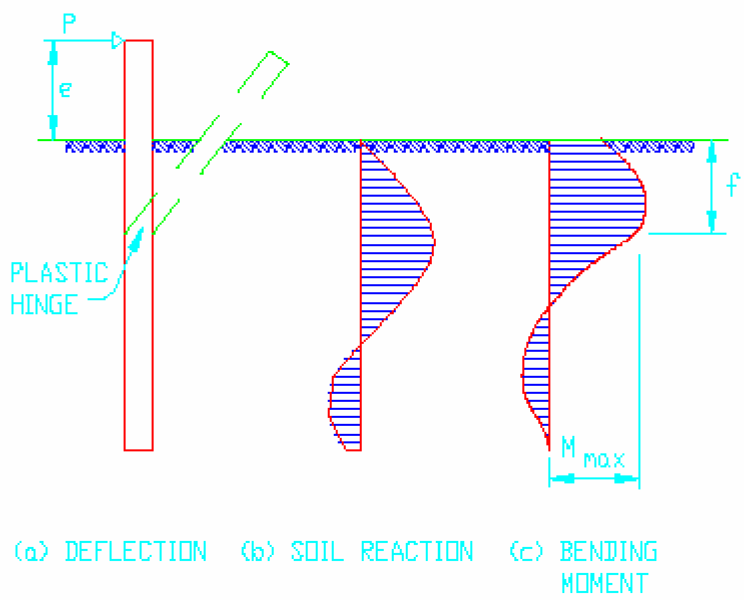
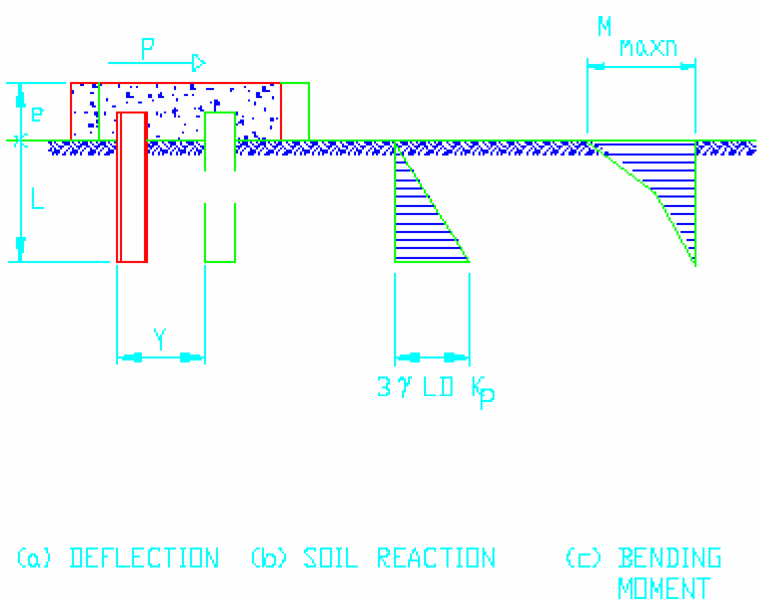


Figure 4: Short Fixed-Headed Pile



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DEFLECTION, SOIL REACTION AND BENDING MOMENT DIAGRAM FOR PILES IN COHESIONLESS SOILS

Figure 5: Intermediate-Length Fixed-Headed Pile

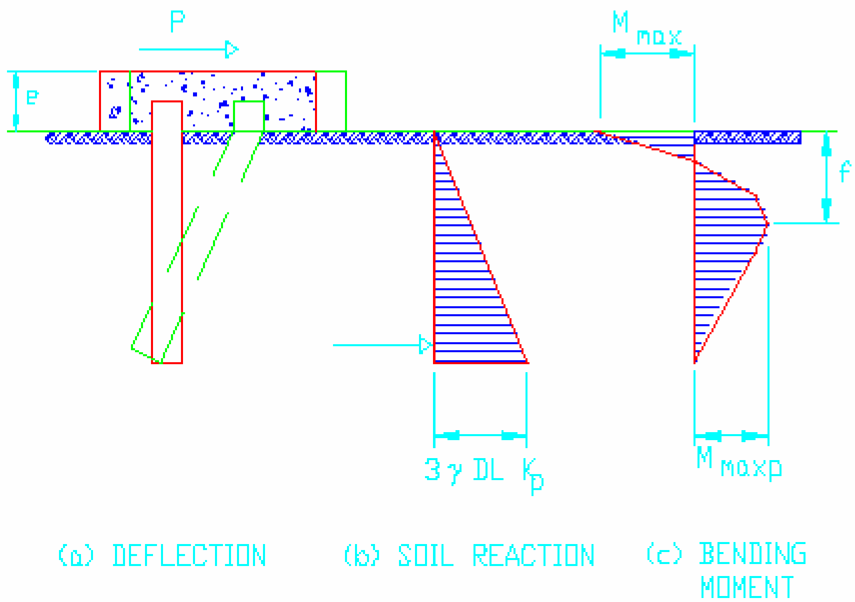
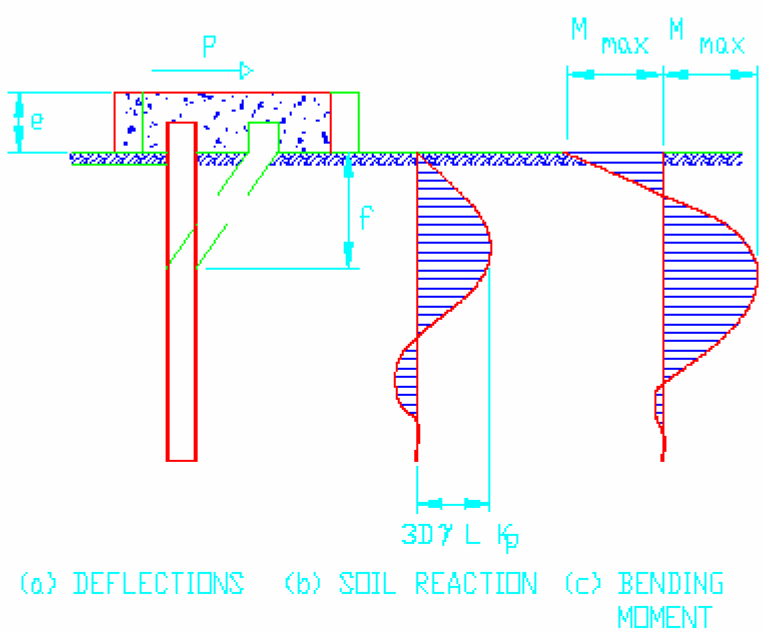


Figure 6: Long Fixed-Headed Pile



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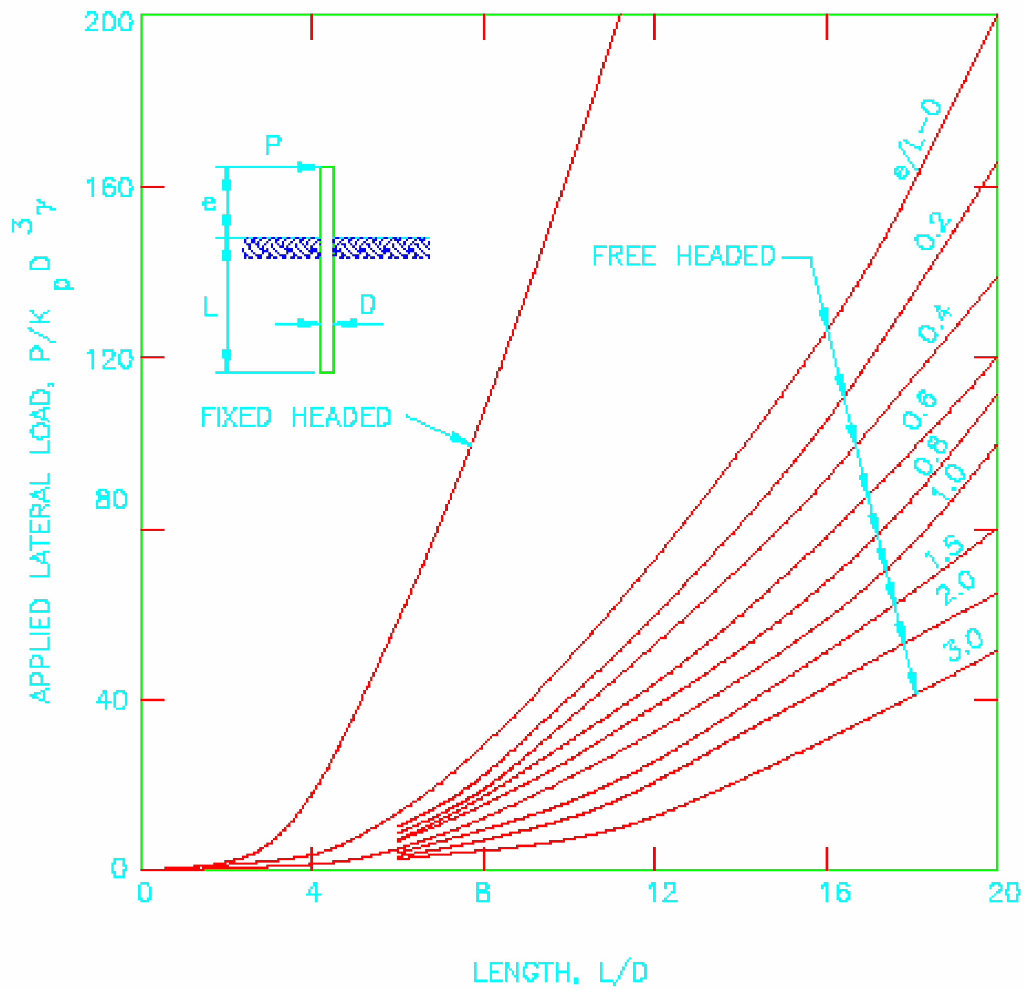
COHESIONLESS SOILS - ULTIMATE LATERAL RESISTANCE OF SHORT PILES

{After Broms (1964)}



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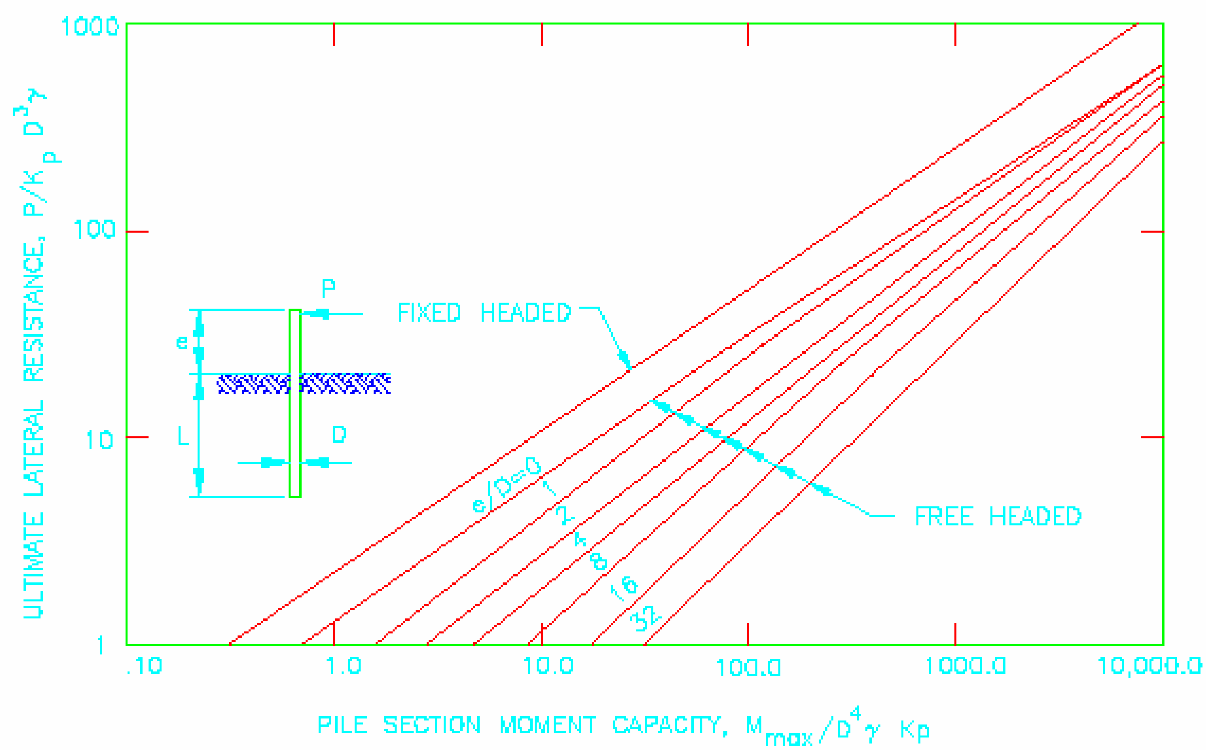
COHESIONLESS SOILS - ULTIMATE LATERAL RESISTANCE OF LONG PILES

{After Broms (1964)}



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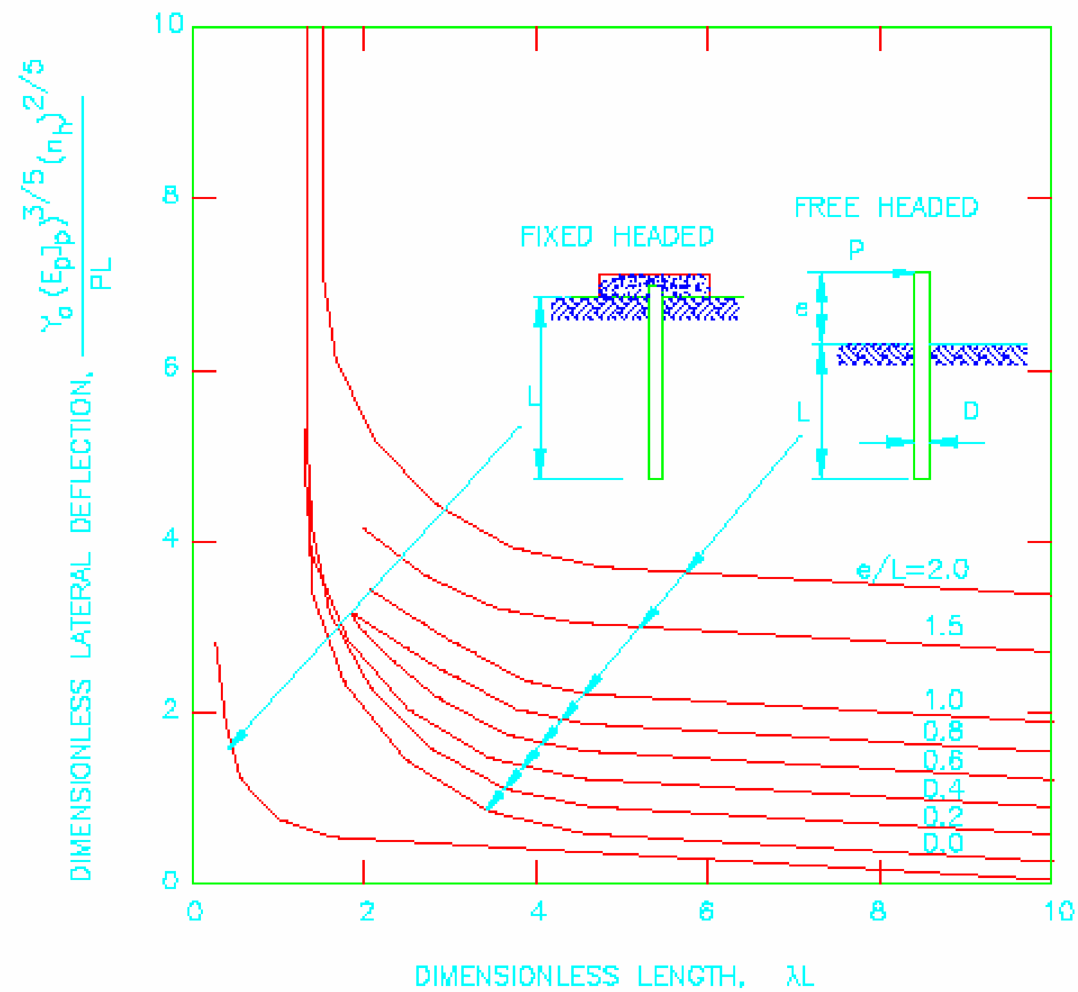
COHESIONLESS SOIL - LATERAL DEFLECTION AT GROUND SURFACE

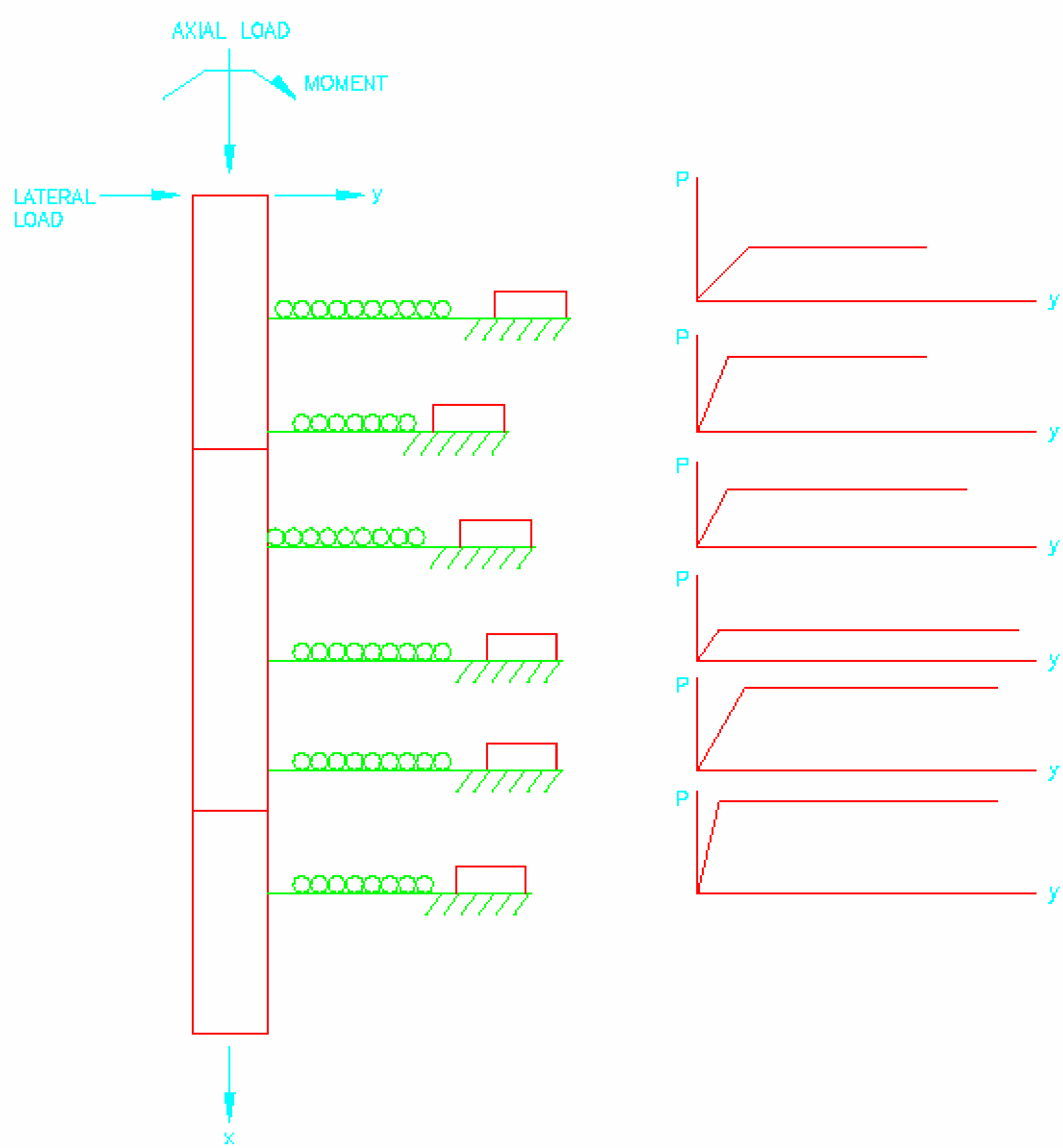
{After Broms (1964)}



FLUOR

MODEL OF A DEEP PILE FOUNDATION UNDER LATERAL LOADING SHOWING CONCEPT OF SOIL RESPONSE CURVES

{Federal Highway Administration (1988)}



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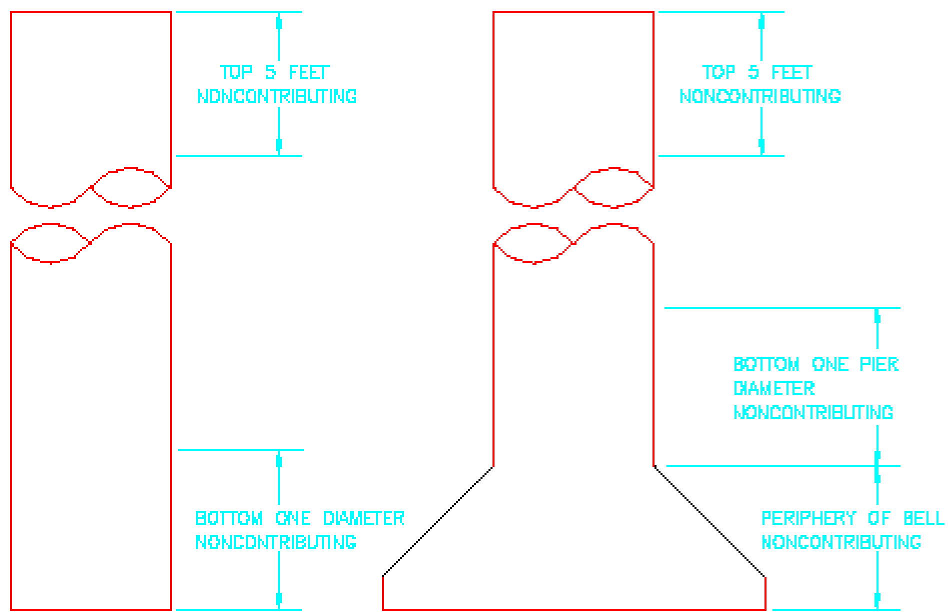
EXPLANATION OF PORTIONS OF DRILLED PIER NOT CONSIDERED IN COMPUTING SIDE RESISTANCE

{Federal Highway Administration (1988)}



STRAIGHT PIER FOUNDATION

BELLED PIER FOUNDATION

RECOMMENDED VALUES OF α FOR DRILLED PIERS IN CLAY		
LOCATION ALONG DRILLED PIER	VALUE OF α_z	LIMITING VALUE OF LOAD TRANSFER, f_{sz} (ksf)
FROM GROUND SURFACE TO DEPTH ALONG DRILLED PIER OF 5 FT *.	0	----
BOTTOM 1 DIAMETER OF THE DRILLED PIER OR 1 PIER DIAMETER ABOVE THE TOP OF THE BELL (IF SKIN FRICTION IS BEING USED).	0	----
ALL OTHER POINTS ALONG THE SIDES OF THE DRILLED PIER.	0.55	5.50

* THE DEPTH OF 5 FEET MAY NEED ADJUSTMENT IF THE DRILLED PIER IS INSTALLED IN EXPANSIVE CLAY, OR IF THERE IS SUBSTANTIAL GROUNDLINE DEFLECTION FROM LATERAL LOADING, OR IF THE FROST DEPTH EXCEEDS 5 FEET.

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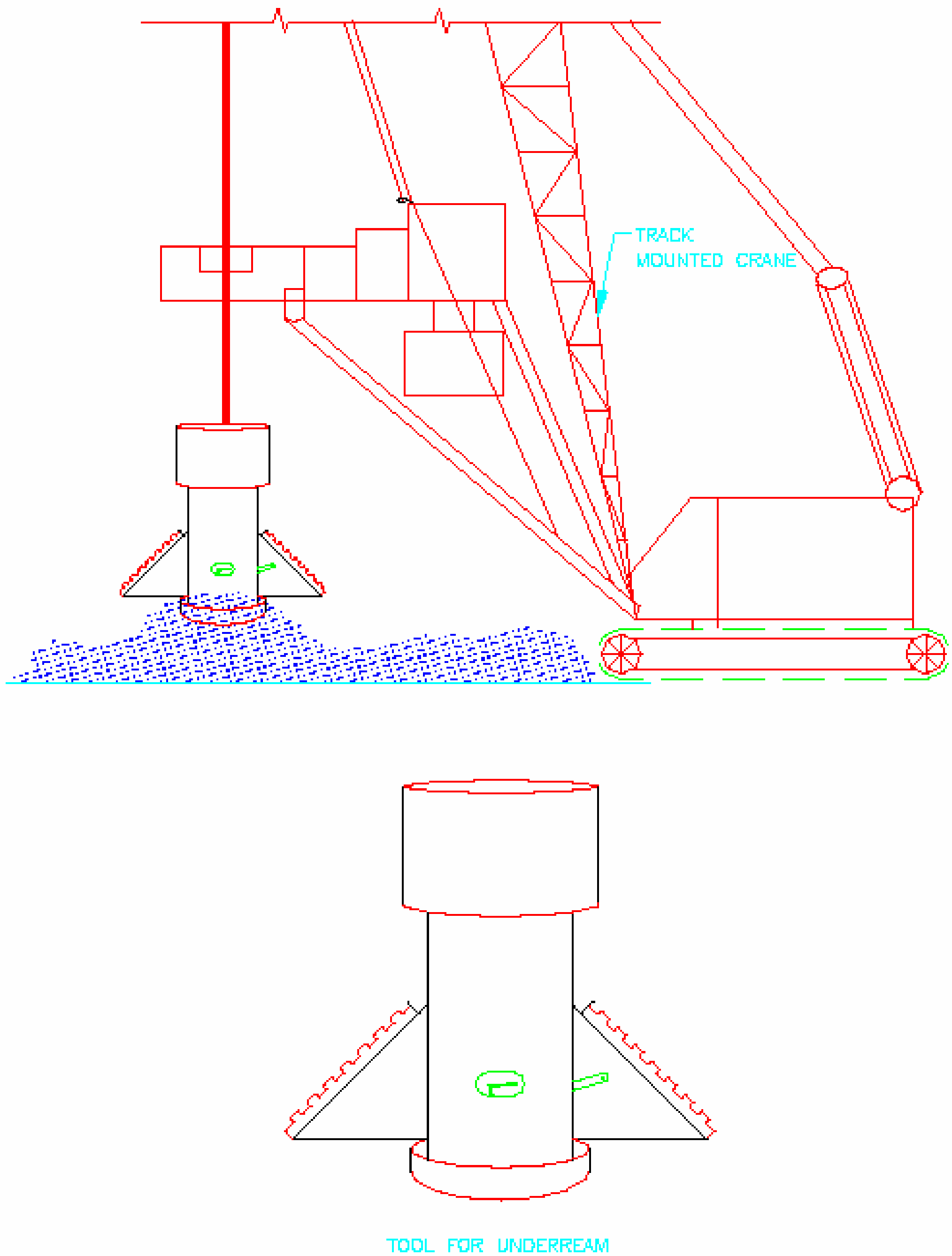
CRANE MOUNTED DRILLING UNIT



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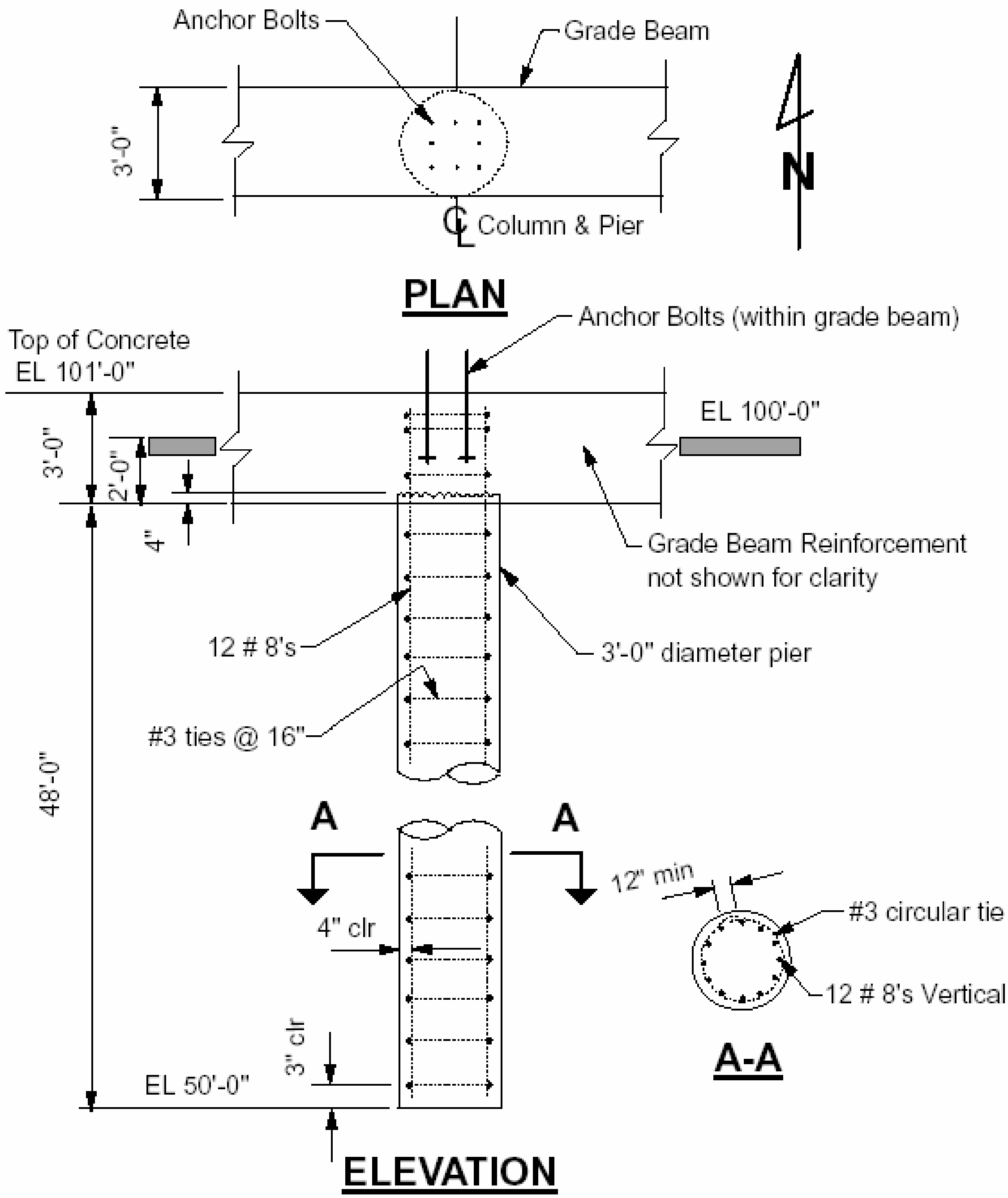
SAMPLE DESIGN 1 : STRAIGHT PILE IN A COHESIVE SOIL



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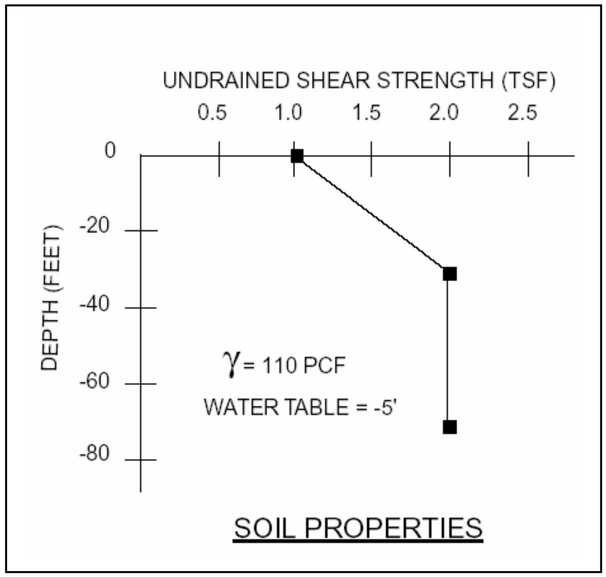
SAMPLE DESIGN 1 : STRAIGHT PILE IN A COHESIVE SOIL

DESIGN DATA

Foundation for a finishing building. Use straight drilled piers with grade beams. Drilled pier spacing is about 10 feet.



Loads:
P_{axial} compression = 280 K (unfactored)
P_{axial} tension = 160 K (unfactored)
Wind shear = 50 K

Note: Axial includes wind loads

Wind axial force is ± 200 K

Axial Compression

Design for a factor of safety 2.5

Group effects will reduce capacity - use efficiency of 0.7
∴ Design load for soil resistance = (280)(2.5) / 0.7

= 1000.0 K

Try 3 FT diameter pile to a depth of 50 FT

(a) Compute side resistance $Q_S = \int_0^L fsz \ dA$

L (FT)	ΔA (SQ FT)	CUZ (KSF)	α	ΔQ(K) (K)
0 - 5	(2.67)(3)π= 25.2	2.16	0	0
5 - 30	(25)(3) π= 235.7	3.16	0.55	409.6
30 – 47	(17)(3) π= 160.2	4	0.55	352.4
47 - 50	(3)(3) π= 28.3	4	0	0
			QS =	762 K

(b) Compute base resistance

C_U (one diameter below base) = 4.0 KSF

$d_c = 1 + 0.2\ L / D = 1 + 0.2(50 / 3) = 4.33 > 1.5$ Use 1.5

$q_b = 9\ C_U$ (Maximum, note depth factor $d_C = 1.5$)

$= 9(4) = 36$ KSF

$A_b = \pi\ (1.5)^2 = 7.07\ FT^2$

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SAMPLE DESIGN 1 : STRAIGHT PILE IN A COHESIVE SOIL

$Q_B = 36(7.07) = 254.5K$

$Q_C = Q_S + Q_B = 762 + 254.5 = 1016.5K > 1000.0K$

3 FT DIA pile, 50 FT embedment, O.K. for compression

Axial Tension

Side resistance, $Q_S = 762K$ (See axial compression calculations)

Weight of pile, $W_c = (5)(7.07)(0.15) + (45)(7.07)(0.09) = 33.9K$

Total ultimate soil = 762 + 33.9 = 7

Resistance, Q_{TT}

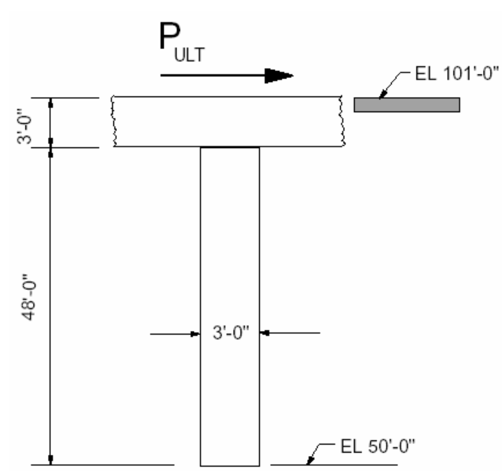
Using global factor of safety of 3, $Q_{allowable\ uplift} = 795.9 / 3 = 265.3K$

Using individual factor of safety, $Q_{allowable\ uplift} = (762 / 4) + (33.9 / 1.5) = 213.1K$

$Q_{allowable\ uplift} = 213.1K > 160.0K$

3 FT DIA pile, 50 FT embedment, O.K. for tension

Lateral Load



The pile is analyzed as fixed headed.

The grade beam provides fixity at top of pier.

Compute ßL

For C_u 1 - 2 TONS, k = 87 LBS/CU IN

For $f_c' = 3000$, E = 3,122,000 LBS/IN ^{2}

$I = p\ r^4 /4 = p(18)^4\ / 4 = 82448$ IN ^{4}

$$\beta = (kD/4EI)^{0.25} \quad = \left[\frac{87(36)}{4(3122000)(82448)}\right]^{0.25}$$

$= 0.007427$

$ßL = 0.007427(48)(12) = 4.28 > 1.5$

$\therefore$ Long pile

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SAMPLE DESIGN 1 : STRAIGHT PILE IN A COHESIVE SOIL

The behavior of a long pile is governed by the pile section capacity. Use ACI column curves to determine pile section capacity. Alternatively, PCA COL (or similar computer program) can be used.

	DEAD		WIND		
For ultimate axial compression =	1.05(80)	+	(1.275)(200)	=	339 K

Allowable ultimate moment capacity = 650 K - FT

For unfactored load, section M_{max} = 650 / 1.275 = 510K - FT

Use an average undrained shear strength at top = (2)(1.33) = 2.67 KSF

For long piles,

$$P_{ult} = \frac{2M_{max}}{1.5D + 0.5f} = \frac{2(510)}{1.5(3) + 0.5f} = \frac{1020}{4.5 + 0.5f}$$

$f = P_{ULT} / 9CD = P_{ULT} / 9(2.67)3 = P_{ULT} / 72.1$

The above equations will result in a quadratic equation.

$36.05f^2 + 324.45f - 1020 = 0 \quad \Rightarrow \quad f = 2.467FT$

$P_{ULT} = 177.89$ K

In lieu of solving the above quadratic equation, curves in attachment 7 could be used.

$M_y / CD^3 = 510 / 2.67(3)^3 = 7.07$ FROM Att. 7 $\Rightarrow \quad P_{ULT} / C_U D^2 = 7$

$\therefore P_{ULT} = 168$ K

Note: The curves are on a log-log scale and will give a solution that is close enough. The quadratic equation solution is accurate.

P_{ULT} = 177.89 K is the ultimate soil lateral resistance developed at which the pile section material reaches its maximum capacity. The maximum available soil resistance is restricted by pile section material capacity. From statics, the load at which soil failure will occur is P_{ULT} = 177.89 K.

The actual allowable lateral load (P_a) requires application of a factor of safety and reduction for group effects.

$P_{allowable} = P_a = P_{ULT}$ / Factor of safety = 177.89 / 2 = 88.95 K

This $P_{allowable}$ is for an isolated pile at 6 to 8 pile diameter spacing.

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SAMPLE DESIGN 1 : STRAIGHT PILE IN A COHESIVE SOIL

For the pile configuration in this example, the group efficiency factor in the longitudinal direction (same as grade beam) is 0.62.

$P_{allowable} = 0.62(88.95) = 55.15$ K > 50 K (actual)

The deflection at grade is (Use formula or Figure 8)

$$Y_o = \frac{P_a \beta}{kD} = \frac{50(1000)(0.007427)}{87(3)(12)} = 0.12 IN < 1/4''$$

3 FT DIA pile, 50 FT embedment, O.K. for lateral load

Concrete Design

Shear - maximum shear occurs at top of pile.

$V_{ult} = 1.275(50) = 63.75$ K

Equivalent square = $\sqrt{\pi (180(18)} = 31.90 IN$

For b = 31.90" d = 27.90"

$V_c = \phi\ 2\ \sqrt{f'_C}\ bd = 0.85(109.5)(31.9)(27.9) / 1000 = 82.85 > 63.75K$

Use # 3 ties at maximum spacing of 16"

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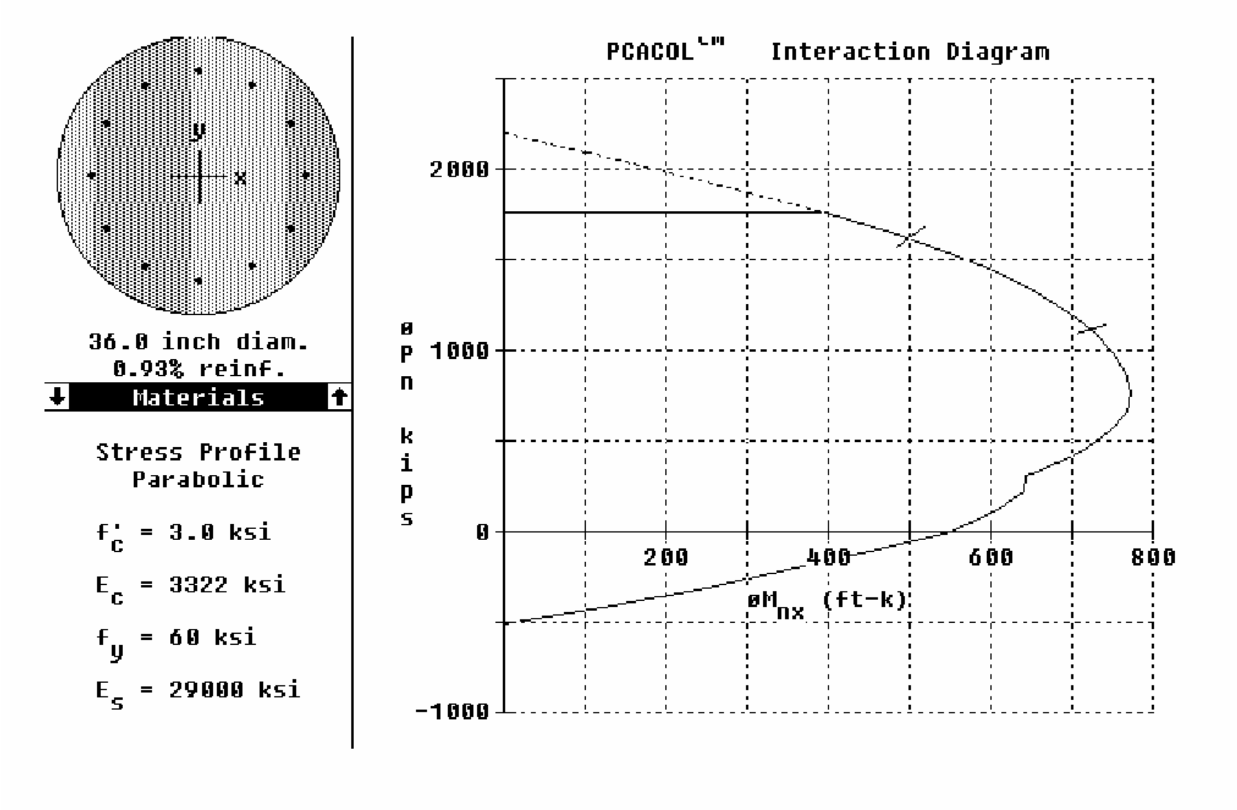
SAMPLE DESIGN 1 : STRAIGHT PILE IN A COHESIVE SOIL

Concrete Design - Pile Section Capacity

PCA COL or equivalent computer program can be used for pile section capacity.

ACI column curves could also be used.



Reinforcement:

12#8
A_{st} = 9.48 in^2
Tied cc=4.00 in
Spacing = 5.99 in
I_x = 82448 in^4
X= 0.00 in
I_y = 82448 in^4
y = 0.00 in

Slenderness not considered x-axis

Also:

ACI 318-89
Reduction: phi c = 0.70
phi b = 0.90
a = 0.80
ε_u = 0.003 in/in

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SAMPLE DESIGN 1 : STRAIGHT PILE IN A COHESIVE SOIL

Concrete Design and Broms Method

Analysis by Broms method requires equilibrium at ultimate soil resistance (failure) and then application of a factor of safety to obtain allowable loads. This method does not have equations for maximum shear and moments in the pile at allowable or other lateral loads. Soil-pile interaction is highly non-linear. To obtain moments and shear for a given applied load requires use of a non-linear computer program.

At $P_{ULT} = 177.89$ K $\Rightarrow M_{max} = 510$ K - FT

The moments in the pile at $P_a = 88.95$ K or $P_{actual} = 50.0$ K is less than pile capacity. This does not mean reduction in reinforcement since pile section capacity determines P_{ULT}.

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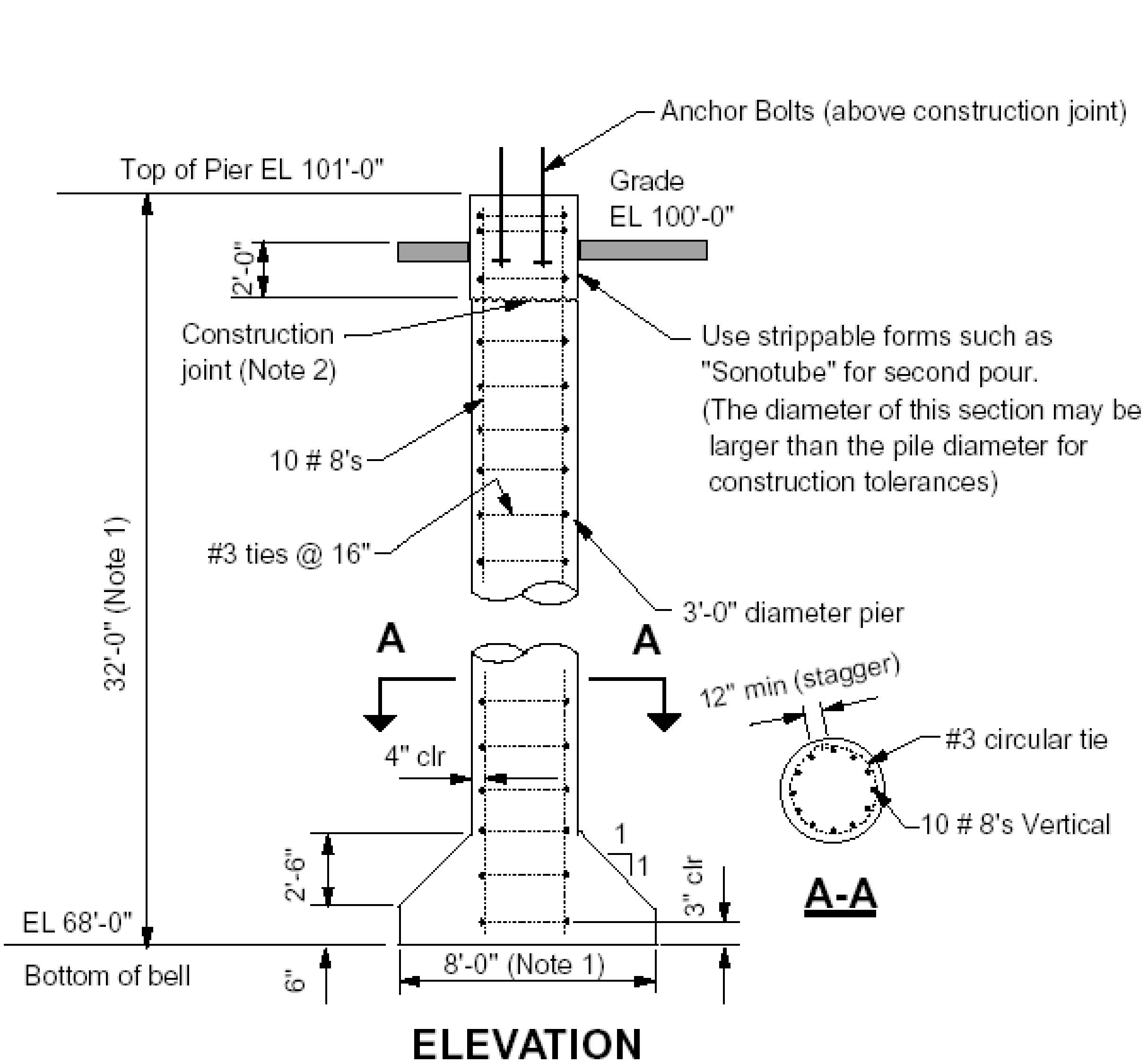
SAMPLE DESIGN 2 : BELLED PIER IN MIXED SOIL PROFILE



Notes:

1. Bells are difficult to form at this depth. Verify with soils consultant and drillingcontractor for bell formability.

2. Construction joint elevation is between 2'-0" to 3'-0" below grade. Vertical bar development length shall be considered.

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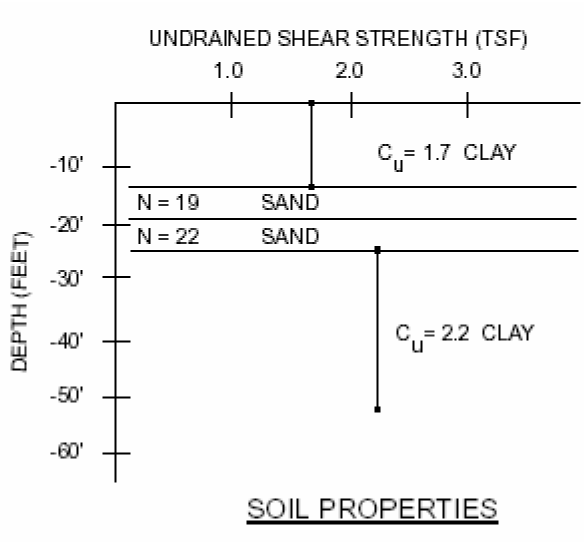
SAMPLE DESIGN 2 : BELLED PIER IN MIXED SOIL PROFILE

DESIGN DATA

A silo structure is supported on drilled pier foundation. The columns are spaced at 10 FT. Belled drilled piers are placed on firm cohesive soil at a depth of 32 FT.



Water table is 6 FT below grade

Soil Profile

0 - 13 FT	Clay $C_u = 1.7$ TON/FT ^{2}
13 - 18 FT	Sand N = 19
18 - 22 FT	Sand N = 22
22 - 50 FT	Clay $C_u = 2.2$ TON/FT 2^2

Soil Density

$\gamma_{Clay} = 110$ PCF 47 PCF (submerged)

$\gamma_{sand} = 115$ PCF 52 PCF (submerged)

Loads:

Dead load = 30 K (empty)

Dead Load = 170 K (operating)

Live load = 20 K

Wind axial = ± 190 K

Wind shear = 25 K

Note!!!

This sample design illustrates computations required for belled pier. The formability of bells at deeper embedment length is difficult. The soils consultant and drilling contractor shall be consulted. For deeper foundations, straight piles shall be preferably used.

Axial Compression:

Maximum axial = 170 + 20 + 190 = 380 K

Design for a factor of safety 3 and group efficiency factor 0.7

$\therefore$ Design load for soil resistance = 380(3) / 0.7 = 1629 K

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SAMPLE DESIGN 2 : BELLED PIER IN MIXED SOIL PROFILE

Try 3' Ø shaft and 8' Ø bell at elevation 32' below grade.

a) Compute soil side resistance (friction)

Cohesive, clay $Q_S = {}_{oL} f_{zs}\, dA$ $\qquad f_{zs} = C_{uz} \int \alpha$

Cohesionless, sand $Q_S = {}_{oL} ß\ z\, dA$ $\qquad ß = 1.5 - 0.135 \int \sigma z$

$1.2 \geq ß \geq 0.25$

For side resistance in sand,

13 - 18' at 15.5' $\quad \sigma_z = (6)(110) + (7)(47) + (2.5)(52) = 1119$ PSF = 1.12 KSF

$ß_z = 1.5 - 0.135\sqrt{\ }\ 15.5 = 0.97$

18 - 22' at 20.0' $\quad \sigma_z = (6)(110) + (7)(47) + (7)(52) = 1353$ PSF = 1.35 KSF

$ß_z = 1.5 - 0.135\sqrt{\ }\ 20 = 0.90$

	Soil Layer	L FT	ΔA SQ. FT	Shear strength or Avg. Eff. Stress	α_z or ß	Δ Q_s K
Clay	0 – 5	5	47.1	3.4	0	0
Clay	5 - 13	8	75.4	3.4	0.55	141
Sand	13 - 18	5	47.1	1.12	0.97	51.2
Sand	18 - 22	4	37.7	1.35	0.9	45.8
Clay	22 – 26	4	37.7	4.4	0.55	91.2
Clay	26 – 32	6	56.5	4.4	0	0
						$Q_S = 329.2$ K

Total side resistance $Q_s = 329.2$ K

(b) Compute base resistance

C_{ub} one diameter below base = 2.2 TONS/FT2 = 4.4 KSF

L = 32 FT

$B_b = 8$ FT

$L/B_b = 32/8 = 4$

$1 + 0.2(L/B_b) = 1.8 > 1.5$

Use 1.5 (Maximum allowed)

$\therefore\ q_b = 9C_{ub} = 9(4.4) = 39.6$ KSF

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SAMPLE DESIGN 2 : BELLED PIER IN MIXED SOIL PROFILE

Since the bell diameter is larger than 75 inches, a reduction factor needs to be applied to q_b.

F_R = Reduction factor = $[2.5 / (aB_B(in.) + 2.5b)]$ $\qquad F_R \leq 1.0$

$a = 0.0071 + 0.0021\ L/B_B$ $\qquad a \leq 0.015$

$b = 0.45\sqrt{C_{ub(ksf)}}$ $\qquad 0.5 \leq b \leq 1.5$

$a = 0.0071 + 0.0021(32 / 8) = 0.0155$ Use 0.015

$b = 0.45\sqrt{4.4} = 0.94$

$F_R = 2.5 / [0.015(96) + 2.5(0.94)] = 0.66$

$q_{br} = F_r q_b = 0.66(39.6) = 26.1$ KSF

$A_b = \pi(4^2) = 50.3\ FT^2$

$Q_B = q_{br} A_b = 26.1(50.3) = 1312.8$ K

Total soil resistance, $Q_{TC} = Q_S + Q_B = 329.2 + 1312.8 =$ $\underline{\underline{1642.0\ K > 1629\ K}}$

Note!!!

The actual soil resistance may be less if a settlement of 1" implied in the above method is not acceptable. The soils consultant recommendations regarding fraction of side and base resistance supersedes above computations.

Axial Tension:

Maximum uplift = 190 - 30 = 160 K

Weight of concrete

Skirt	$0.5(\pi(4^2)(0.09)$	$= 25.14(0.09) = 2.26$
Bell	$1/3\pi(2.5)(8^2/4 + 3^2/4 + 8(4/4))(0.09)$	$= 68.73(0.09) = 6.19$
Submerged Pier	$\pi(1.5^2)(23)(0.09)$	$= 162.6(0.09) = 14.63$
Dry Pier	$\pi(1.5^2)(7)(0.15)$	$= 49.5(0.15) = \underline{\underline{7.42}}$
		30.50

Weight of soil above bell

Skirt	0	= 0
Bell	$(\pi(4^2)(2.5) - 68.7))(0.05)$	= 2.85

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SAMPLE DESIGN 2 : BELLED PIER IN MIXED SOIL PROFILE

Submerged Pier$(\pi(4^2)(23) - 162.6))(0.05)$ = 49.68

Dry Pier $(\pi(4^2)(6) - 42.4))(0.11)$ = <u>28.52</u>

81.05 K

Soil side resistance of a fictitious cylinder same diameter as bell

$Q_{Bell} = (8' / 3')(329.2\text{ K}) = 877.87\text{ K}$

↑ (Total side resistance).

Using a global factor of safety of 3,

$Q_{allowable\ uplift} = (30.5 + 81.05 + 877.87) / 3 = 989.4 / 3 = 329.8\text{ K}$

Using individual factor of safety,

$Q_{allowable\ uplift} = (877.87 / 4) + (30.5 + 81.05)/ 1.5 = 219.47 + 74.37 = 293.8\text{ K} \leftarrow$

$Q_{allowable\ uplift} = \underline{293.8 > 160\text{ K}}$

Pier O.K. for uplift

<u>Check Unreinforced Bell Concrete Capacity for Axial</u>

ACI calls out for a minimum bell angle of 55°. This requires use of a 60 series reamer which is not desirable for large diameter piles. Also, a 55° bell requires more concrete quantity.

Use of a 45° bell is permitted provided plain Unreinforced bell or underream has adequate strength.

For details, refer to:

"Plain Concrete Underreams for Drilled Shafts",

by J. S. Farr and L.C. Reece,

ASCE Structural Journal ST 6 June 1980.

Ultimate axial compression = 1.05(170) + 1.275(20 + 190) + 446.3 K

Soil side resistance = 329.2 K (From Page 3)

At base, ultimate axial load = 446.3 - 329.2 = 117.1

Ultimate weight of conc + soil = 1.05(30.5 + 81.05) = 117.1

q_{ULT} @ base = $(117.1 + 117.1) / (\pi\ 4_2) = 4.66$ KSF < 5 KSF allowed for a 45° bell.

Refer to figure 12 of above reference. ∴ 45° bell is adequate

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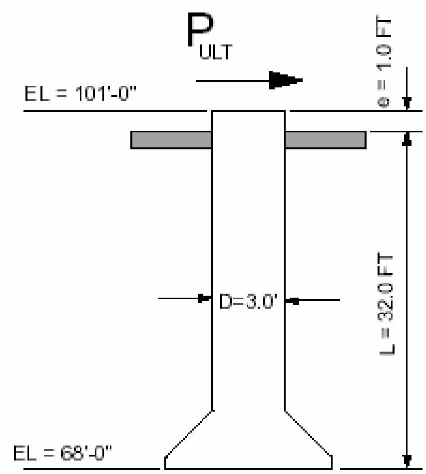
SAMPLE DESIGN 2 : BELLED PIER IN MIXED SOIL PROFILE

Lateral loads:

The pile is analyzed as free headed. Pile behavior is governed by soil conditions at the top. For this case, cohesive (clay) soil equations are used.

$C_u = 1.7$ TSF = 3.4 KSF



Compute ßL

For Cu 1.2 TSF, k = 87 Lbs/IN^3

For f_c' = 3000, E = 3122000 Lbs/IN^2

$I = \pi\ ^4/4 = \pi(18)^4 / 4 = 82448\ IN^4\ \tau$

$$ß = (kD / 4EI)^{0.25} = \left[\frac{87(36)}{4(3122000)(82448)}\right]^{.25}$$

= 0.007427

ßL = 0.007427(32)(12) = 2.85 > 2.5

∴ Long pile

The behavior of the long pile is governed by pile section capacity. Use ACI column curves or PCA COLS computer program as shown on SHT 8:

Axial P_u = 1.05(170) + 1.275(20 + 190) = 446.3 K

Allowable ultimate moment capacity = 660 FT-K

For unfactored load, section M_{max} = 660 / 1.275 = 517 FT-K

For long piers,

$$P_{ULT} = \frac{M\max}{e+1.5+0.5f} = \frac{517}{1+1.5(3)+0.5f} = \frac{517}{5.5+0.5f}$$

$$f = \frac{P_{ULT}}{9CD} = \frac{P_{ULT}}{9(3.4)(3)} = \frac{P_{ULT}}{91.8}$$

The above equations results in a quadratic equation

$45.9f^2 + 504.9f - 517 = 0$ f = 0.94 FT ⇒

P_{ULT} = 86.6 K

In lieu of solving the above quadratic equation, curves in Figure 7 could be used.

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SAMPLE DESIGN 2 : BELLED PIER IN MIXED SOIL PROFILE

$$M_y / CD^3 = 517 / (3.4x3^3) = 5.63 \rightarrow P_{ULT} / CD^2 = 2.7 \rightarrow P_{PULT} = 82.6$$

P_{ULT} = 86.6 K is the ultimate soil lateral resistance developed along the pile length when the pile section material reaches its maximum capacity. The maximum available soil resistance is restricted by pile section material capacity.

P_{ULT} = 86.6 K is the load at which soil failure will occur.

The actual allowable lateral load (P_a) requires application of a factor of safety and reduction for group effects.

Using a factor of safety of 2 and a group efficiency or reduction factor of 0.62,

$$P_{allowable} = P_a = (86.6 / 2) x0.62 = 26.85 \text{ K} > 25.0 \text{ K (actual)}$$

The deflection at grade for actual lateral load is (Figure 8):

$$Y_O = \frac{2p\beta(e\beta + 1)}{kD} = \frac{2(25000)(0.007427)(0.007427 + 1)}{87(36)} = 0.12IN < 1/4IN$$

3' Ø pile, 32 FT embedment is adequate for applied lateral loads.

Concrete Design - Lateral Loads:

Shear - maximum shear occurs at top of pile

V_{ULT} = 1.275(25) = 31.88 K

Equivalent square = $\sqrt{\pi 18^2}$ = 31.90 IN

For b = 31.90 IN

d = 27.90 IN

$V_c = Ø\ 2\ bd\ f_c$

V_c = 0.85(109.5)(31.9)(27.9) / 1000 = 82.85 K > 31.88 K

Use #3 ties at maximum spacing of 16 IN

Moment - The actual moment in the pile at P_a = 26.85 or P = 25.0 is less than the pile capacity (at P_{ULT} = 86.6,

M_{max} = 517 FT-K). For additional discussion, see sheet 8 (Concrete Design and Broms Method).

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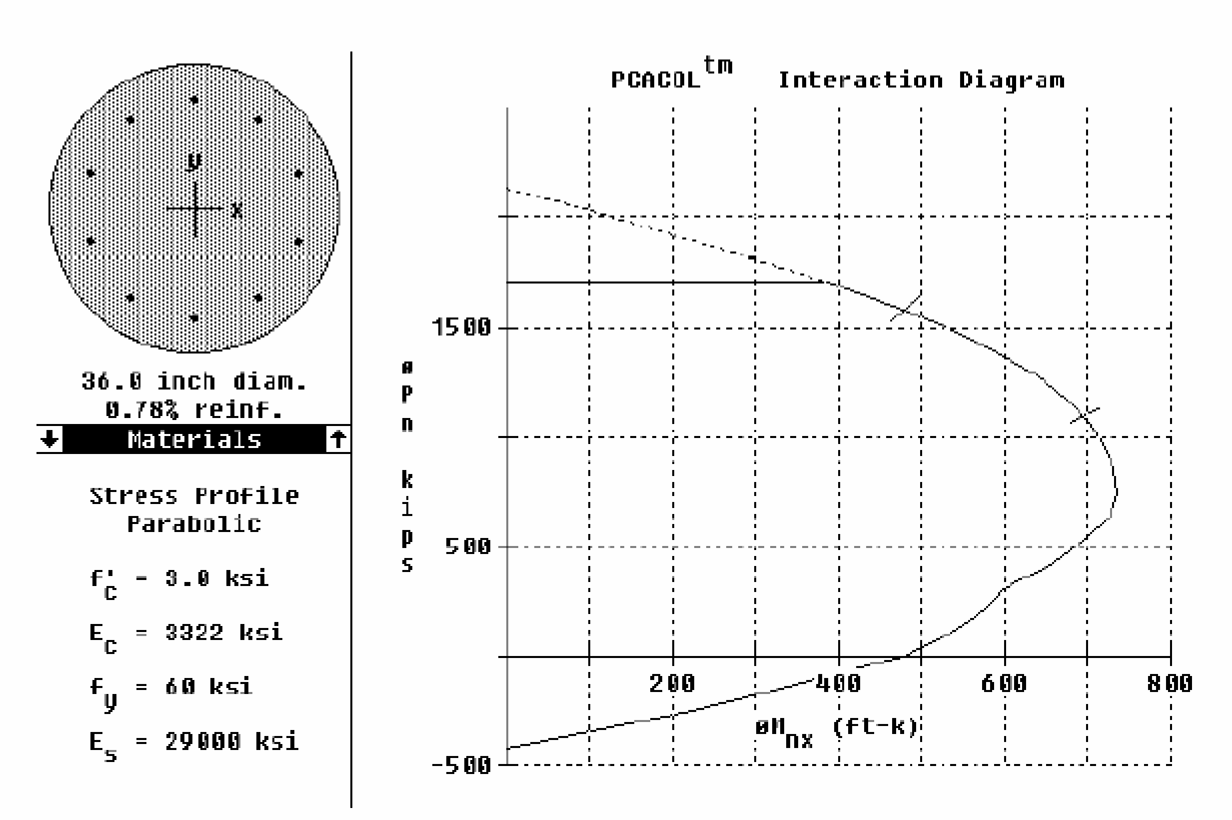
SAMPLE DESIGN 2 : BELLED PIER IN MIXED SOIL PROFILE

Concrete Design - Pile Section Capacity

PCA COL or equivalent computer program can be used for pile section capacity.
ACI column curves could also be used.



Reinforcement:
- 10#8
- A_{st} = 7.90 in^2
- Tied cc=4.00 in
- Spacing = 7.34 in
- I_x = 82448 in^4
- $\overline{X}$ = 0.00 in
- I_y = 82448 in^4
- $\overline{Y}$ = 0.00 in

Slenderness not considered x-axis

Also:
- ACI 318-89
- Reduction phi c=0.70
- phi b=0.90
- a = 0.80
- ε_u = 0.003 in/in

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SAMPLE DESIGN 2 : BELLED PIER IN MIXED SOIL PROFILE

Concrete Design and Broms Method

Analysis by Broms Method requires equilibrium at ultimate soil resistance (failure) and then application of a factor of safety to obtain allowable loads. This method does not have equations for maximum shear and moments in the pile at allowable or other lateral loads. Soil-pile interaction is highly non-linear. To obtain moments and shear for a given applied load requires use of a non-linear computer program.

At $P_{ULT} = 86.6$,

$M_{max} = 517$ FT-K

The moments in the pile at $P_a = 26.85$ K or $P_{actual} = 25.0$ K is less than pile capacity. This does not mean reduction in reinforcement since pile section capacity determines P_{ULT}.

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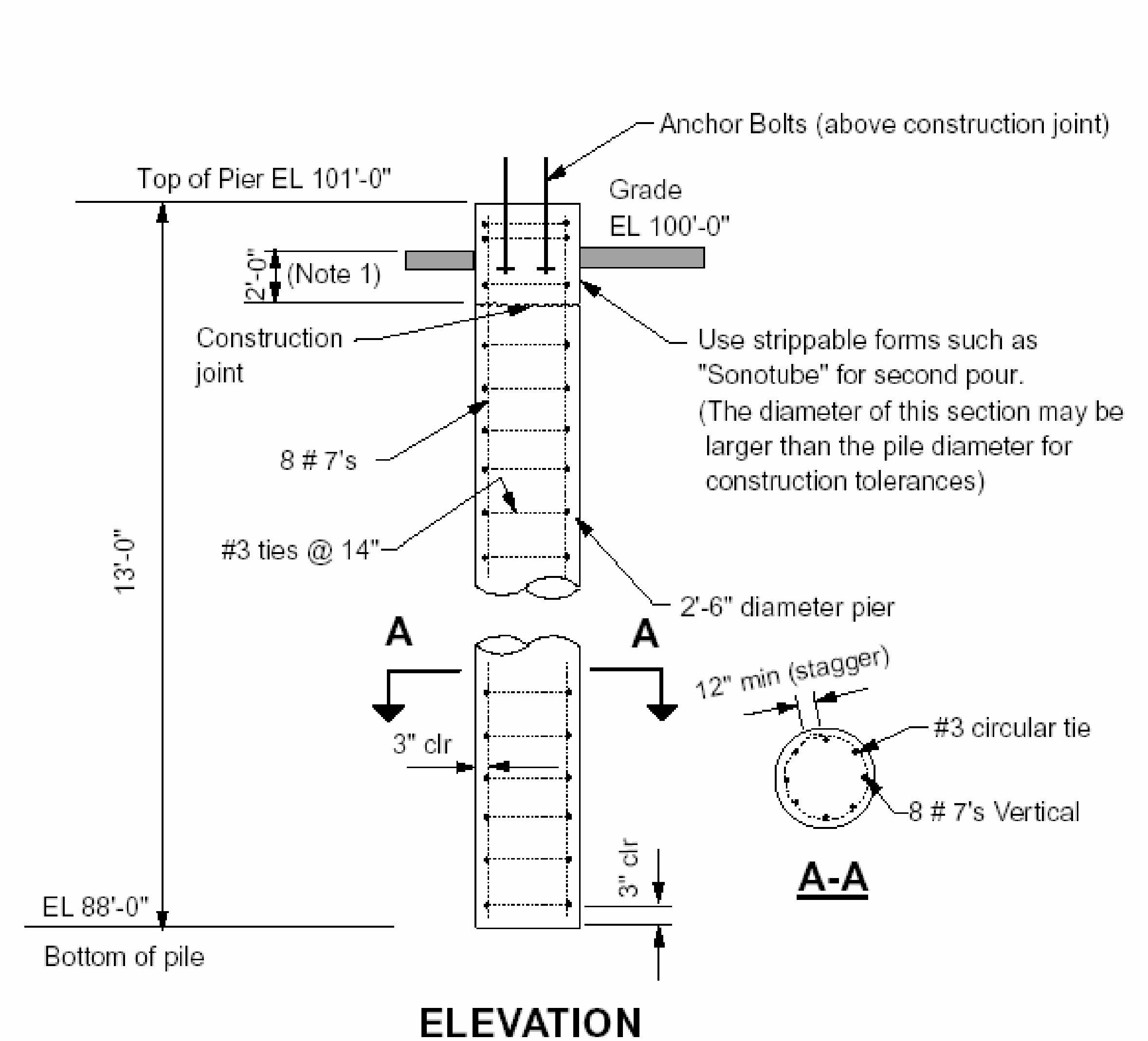
SAMPLE DESIGN 3: SHORT STRAIGHT PILE IN SAND



Notes:

1. Construction joint elevation is between 2'-0" to 3'-0" below grade. Vertical bar development length, anchor bolt length,final grade, etc. shall be considered.

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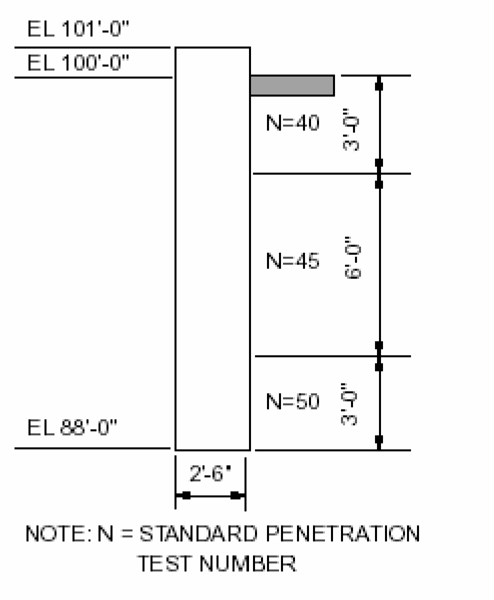
SAMPLE DESIGN 3: SHORT STRAIGHT PILE IN SAND

DESIGN DATA

This example illustrates use of straight piles in cohesionless soil (sand).



Soil Profile

Dense sand (30 < N < 50)

γ = 120 Lbs/FT^3

Q = 37°

Water table is 16FT below grade

Note: N = Standard penetration test number

Loads:

The column has a pin base support.
Dead load = 80 K (empty)
Wind axial = ± 40 K
Wind shear = 14 K

Note!!!

The soil report will provide recommendations for suitability of drilled pier foundations in cohesionless soils. Belled piers are difficult to form in sand. Shallow straight piles in sand are normally not recommended.

Axial Compression:

Maximum compression, Dead + Wind = 80 + 40 = 120 K

Design for a factor of safety of 3 against ultimate soil resistance.

Design compression - 3(120K) = 360 K

Try 2' 6 " Ø pile, embedment 12 FT

Total soil resistance = side resistance + base resistance

$$Q_{TC} = Q_S + Q_B$$

a) Side resistance $Q_S = \int_0^L ß\sigma_Z \, dA$

$$ß = 1.5 - 0.135\sqrt{Z}$$

$$1.2 \geq ß \geq 0.25$$

ß is a factor dependent on depth.

σ_Z average stress at depth Z.

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SAMPLE DESIGN 3: SHORT STRAIGHT PILE IN SAND

Depth FT	ΔA FT^2	Avg. Stress, σ_Z	ß (MAX 1.2)	ΔQ (K)
0 – 3	π (2.5)(3) = 23.6	0.12(1.5) = 0.18	$1.5 - 0.135\sqrt{1.5} = 1.33$	5.1 K
3 – 9	π (2.5)(6) = 47.1	0.12(6.0) = 0.72	$1.5 - 0.135\sqrt{6.0} = 1.17$	39.7 K
9 - 12	π (2.5)(3) = 23.6	0.12(10.5) = 1.26	$1.5 - 0.135\sqrt{10.5} = 1.06$	31.5 K
			Total side resistance	76.3 K

(b) Base resistance $Q_B = A_b q_b$

$q_b = 1.2$ N for 0 N < 75 ≤

N one diameter below base is 50

$q_b = 1.2 N = 1.2(50) = 60$ KSF

$A_b = \pi\ 1.25^2 = 4.91\ FT^2$

$Q_B = 4.91(60) = 294.6$ K = total base resistance

(c) Total ultimate soil resistance

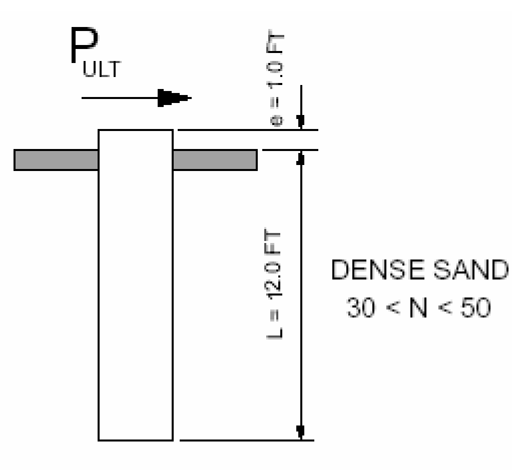
$Q_{tc} = Q_s + Q_B = 76.3 + 294.6 = 370.9$ kips > 360 kips

30" pile embedded 12'-0" is adequate for compression ϕ

Axial Tension:

There is no uplift.

Lateral loads:



The pile is analyzed as free headed.

Compute dimensionless length, ZL

For dense sand above water table,

$\eta_h = 56\ TONS/FT^3 = 65\ Lbs/IN^3$

For $f_c' = 3000$

$E_p = 3122000\ Lbs/IN^2$

For 30" Ø pile, $I_p = \pi r^4/4 = \pi\ 15^4/4 = 39761\ IN^4$

$$\eta = (n_h/E_pI_p)^{0.2} = \left[\frac{65}{3122000(39761)}\right]^{0.2} = 0.0139\ IN^{-1}$$

$\eta L = 0.0139(12)(12) = 2.0 \leftarrow$ Treat as short pile

$K_p = (1 + \sin Ø)/(1 - \sin Ø) = (1 + \sin 37) / (1 - \sin 37) = 1.6 / 0.398 = 4.02$

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SAMPLE DESIGN 3: SHORT STRAIGHT PILE IN SAND

$$P_{ULT} = \frac{\gamma D L^3 K_P}{2(e+L)} = \frac{0.12(2.5)(12^3)(4.02)}{2(1+12)} = 80.15\text{K SAY 80K\{Alternatively, use flg.10\}}$$

$$\text{f}=0.186\,\Delta\left[\frac{P_{ULT}}{\gamma D K_P}\right] = 0.186\left[\frac{80.15}{0.12(2.50(4.02)}\right]^{\frac{1}{2}} = 435.4FT-K$$

$M_{maxp} = P_{ULT}$ (e + f) - P_{ULT} (f / 3) = 80.15 (1 + 6.65) - 80.15 (6.65 / 3) = 435.4 FT-K

g = L - F = 12 - 6.65 = 5.35 FT

The actual allowable lateral load (P_a) requires application of a factor of safety and reduction for group effects. Furthermore, in cohesionless soils (sand), deflection at grade could restrict allowable load.

$P_{allowable} = P_a = P_{ULT}$ / Factor of safety = 80 / 2 = 40 > 14 K

The deflection at grade is (use formula or figure 12)

$$Y_o \text{ @ } 14\text{ K} = \frac{18P(1+1.33e/L)}{L^2\eta_h} = \frac{18(14)(1000)[1+(1.33/12)]}{(12x12)^2(65)} = 0.21\text{ IN} < 1/4\text{ IN}$$

$\therefore$ O.K.

<u>Concrete Design - Lateral Loads:</u>

A) Moments

At a lateral load of 80 K, the soil resistance along the pile length will develop its maximum ultimate capacity i.e. at this load, the soil will fail.

The maximum moment (unfactored) in the pile section at a lateral load of 80 K is 435.4 FT-K.

Analysis by Broms Method requires eqilibrium at ultimate soil resistance (failure) and then application of a factor of safety. Broms Method does not have equations for the maximum moments in the pile at allowable lateral load. The soil pile interaction between P_a and P_{ULT} is highly non-linear. The actual pile moment at allowable lateral load, P_a, will require use of a non-linear computer program.

Conservativley, assume that the maximum moment at P_a is the same as at P_{ULT} i.e. M_{maxp} = 435.4FT-K. (Note – this assumption does not result in excessive reinforcement. In most cases, 1/2 to 1% reinforcement will develop the above moment capacity). For a lateral load less than P_a, linear interpolation is permitted. At lateral loads less than 1/3 to ½ P_{ULT}, soil pile interaction is approximately linear.

maximum moments in the pile at P = 14 K = (14 / 40) 435.4 = 152.4 FT-K

Ultimate moment in pile = 1.275(152.4) = 194.3 FT-K
↑ lateral load is due to wind.

Ultimate axial load in pile = 1.05(D) + 1.275 (W)

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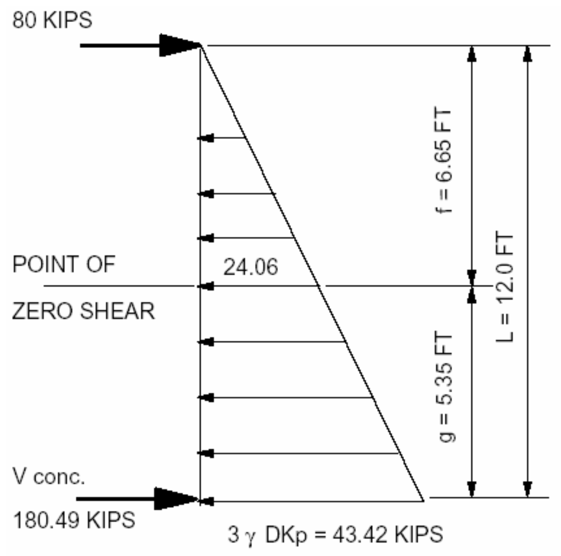
SAMPLE DESIGN 3: SHORT STRAIGHT PILE IN SAND

$= 1.05 (80 + \pi (1.25^2) (6.65)(0.15))+ 1.275(40) \qquad = 140.1$ K

weight of pile is negligible

Using ACI column curves or PCA COL computer program generated curves (sheet 6), 8 # 7 (0.7% reinforcement) is adequate.



SOIL REACTION AT FAILURE

B) Shear

For an applied load $P_{ULT} = 80$ K

Concentrated load at bottom is:

$V_{conc} = 1.5\ \gamma DKpg\ (f + L)$

$= 1.5\ (0.12)(2.5)(4.02)(5.35)(6.65 + 12)$

$= 180.49$ K

@ $P_{ULT} = 80$ K, the soil will reach its ultimate capacity and fail.
As discussed on Sht. A, Broms Method provides shear along the pile at failure. This method does not have equations for shear at allowable loads. The discussion for shear is similar to moment on Sht. A.

Shear at bottom of pile = (14 / 40)180.49 = 63.17 K

@ $P_a = 14$ K

$V_{ULT} = 1.275(63.17) = 80.5$ K

For 30" Ø, equivalent square = 26.58 IN

$b_o = 26.58"$

$d = 23.58"$

$V_c = Ø\ (2) \sqrt{f_c^i}\ b_o d = 0.85\ (109.5)(23.58)(26.58) / 1000 = 58.33$ K

$V_s = 80.5 - 58.33 = 22.17$ K

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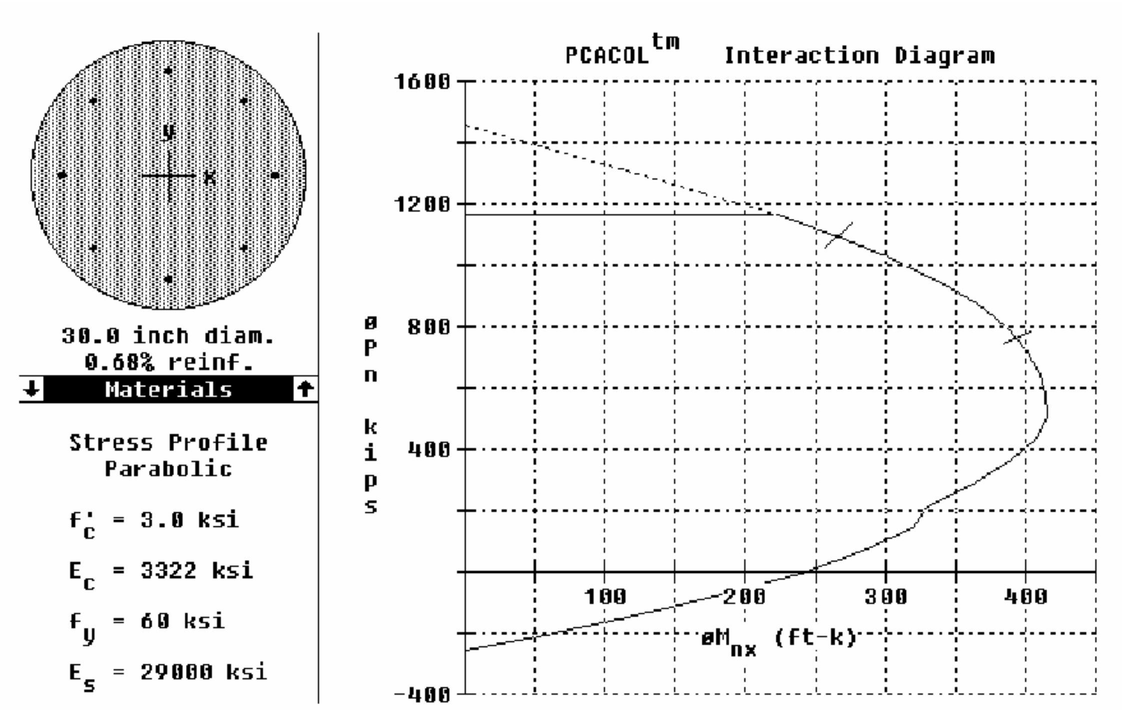
SAMPLE DESIGN 3: SHORT STRAIGHT PILE IN SAND

For #3, S = A_v fyd / V_s = 2(0.11)(60)(23.58) / 22.17 = 14.04 IN

16 longitundinal bar diameters is 14 INS

Use #3 circular ties @ 14 IN

Pile Section - Concrete Capacity



Reinforcement:
- 8#7
- A_{st} = 4.80 in^2
- Tied cc=3.00 in
- Spacing = 7.97 in
- I_x = 39761 in4
- $\overline{X}$ = 0.00 in
- I_y = 39761 in4
- $\overline{Y}$ = 0.0 in
- Slenderness not considered x-axis

Also:
- ACI 318-89
- Reduction phi c=0.70
- phi b=0.90
- a = 0.80
- ε_u =0.003 in/in

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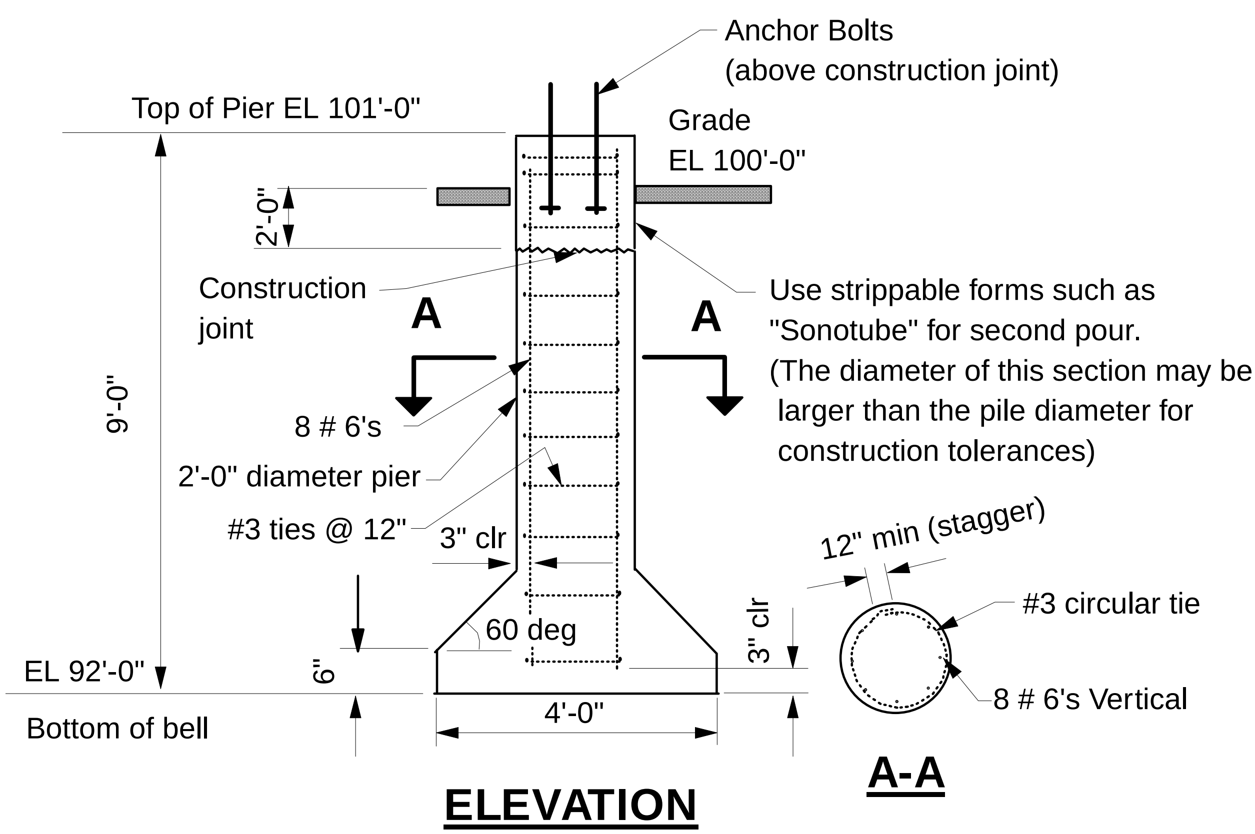
SAMPLE DESIGN 4: SHORT BELLED PIER IN CLAY



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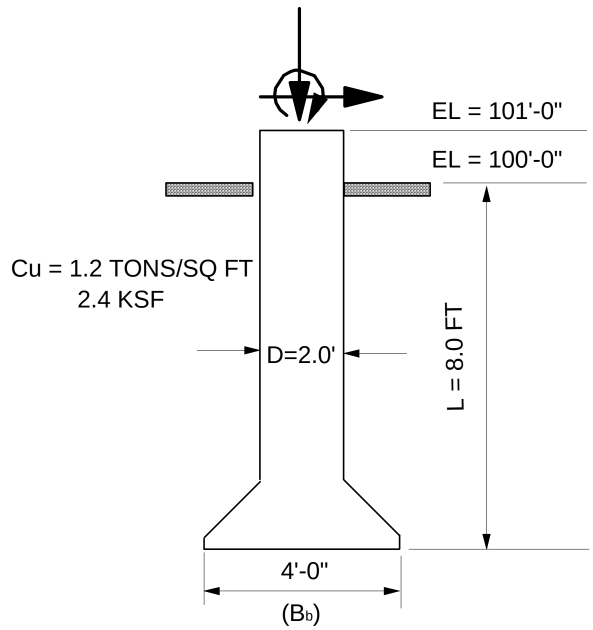
SAMPLE DESIGN 4: SHORT BELLED PIER IN CLAY

DESIGN DATA

This example illustrates use of short (rigid) belled piers in moderately stiff clays.



Soil Properties:

Soil undrained shear strength 1.2 TSF

Water table is 6 FT below grade

Soil density 110 Lbs/FT3

Design Loads:

Dead loads = 40 K
Wind axial = 35 K

Wind shear = 5 K @ EL 101' 0"
Wind moments = 25 K @ EL 101' 0"

Note!!
The soil report will provide recommendations for allowable loads versus settlements. The method presented here for axial load is a general method that will yield a short term settlement of approximately 1 inch. This settlement may be untolerable. A reduced allowable load may be required.

Some sensitive piping systems may restrict settlements. Group effects will also increase settlement and reduce allowable axial capacity.

Axial Compression:

Maximum compression = dead + wind = 40 + 35 = 75 K

Use a factor of safety of 3 against ultimate soil resistance.

Design compression = (3)(75) = 225 K

(a) Side resistance

None. The embedment length is very short.

(b) Base resistance

This method results in a short term settlements of approximately 1 inch.

S_c Shape factor = 1.2

d_c = Depth factor = $1 + 0.2\ L / B_b \quad L / B_b\ \ 2.5$

$d_c = 1 + 0.2(8 / 4) = 1 + 0.4 = 1.4 < 1.5$ (maximum depth factor)

$q_b = 5.14\ C_u\ S_c\ d_c\ 9\ C_{ub}$

$q_b = 5.14(1.2)(1.4)C_{ub}$

$= 8.64C_{ub}$

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SAMPLE DESIGN 4: SHORT BELLED PIER IN CLAY

=8.64((1.2)(2)) = 20.74 KSF   Soil ultimate end bearing

$A_b = \pi\ (2^2) = 12.56\ FT^2$

$Q_B = A_b\ q_b$ 12.56(20)(74) = 260.5 K > 225 K (design)

4 FT diameter bell is adequate.

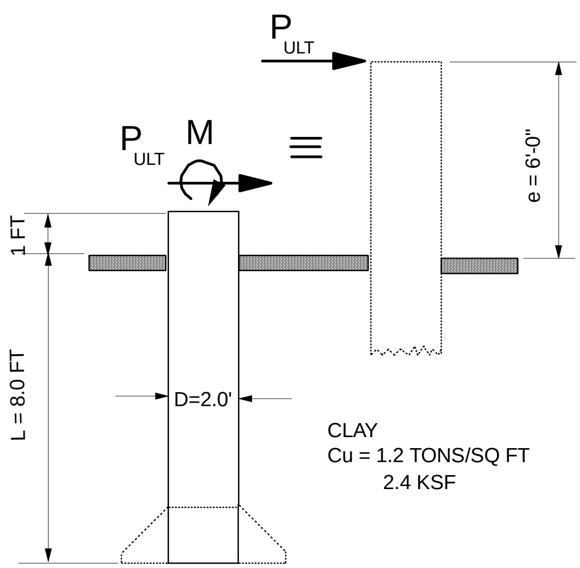
Axial Tension:

There is no uplift.

Unreinforced Concrete Bell:

60° bell is used. If a 45° bell is used check concrete capacity. "Plain Concrete Underreams for Drilled Shafts" J.C. Farr and L.C. Reese, ASCE St6, June 1980.

Lateral Load:



The pile is analyzed as free headed.

Moment = 25 FT-K @ 1 FT above grade

This is equivalent to a 5 K shear

At 5 FT above top of pier

∴ Shear = 5 K

e = 6 FT

Compute ßL

For C_u 1.2 TONS, k = 87 Lbs/IN^3

For f_c’ = 3000, E_p = 3122000 Lbs/IN^2

For 24" Ø pier, $I_p = \pi\ \tau^4/4$ = 16288 IN^4

$$\beta = \left[\frac{kD}{4E_p I_p}\right]^{0.25} = \left[\frac{87(24)}{(3122000)(16288)}\right]^{0.25}$$

= 0.01423 IN-1

ßL = 0.01423(8)(12) = 1.37 < 1.5   ... Short pile

For short pile,

$f = P_{ULT} / 9C_uD = P_{ULT} / (9x2.4x2) = P_{ULT} / 43.2$

$M_{maxp} = P_{ULT}\ (e + 1.5D + 0.5f) = P_{ULT}(6 + 1.5(2) + 0.5f) = P_{ULT}(9 + 0.5f)$

M_{maxp} = 2.25 C_uDg2 = 2.25(2.4)2g2 = 10.8g2

L = 1.5D + f + g

g = 8 - 1.5(2) - f = (5 - f)

The above results in a quadratic equation in f:

10.8f 2 + 496.8f - 270 = 0 f = 0.537 FT

g = 4.46 FT

SAMPLE DESIGN 4: SHORT BELLED PIER IN CLAY

P_{ULT} = 23.2 K

M_{maxp} = 214.8 FT-K

In lieu of solving the above quadratic equation, curves in figure 6 could be used. The curves are difficult to interpolate at the lower range.

L / D = 8 / 2 = 4

e / D = 6 / 2 = 3 $P_{ULT} / C_u D^2 = 2.2$

P_{ULT} = 2.3(2.4)(22) = 22.1 K

The ultimate soil resistance developed along the pile length is PULT = 23.2 K. The soil resistance developed depends upon the eccentricity of applied lateral load; i.e. moments at top of pile.

The actual allowable lateral load (Pa) requires application of a factor of safety and reduction for group effects. Also, the deflection at grade is a limiting factor in restricting allowable lateral load.

$P_{allowable} = P_a = P_{ULT}$ / factor of safety = 23.2 / 2 = 11.6 K > 5 K

The deflection at grade is (use formula or figure 8 for small e /L)

$$Y_o \; @\, 5K = \frac{4Pa(1+1.5e/L)}{kDL} = \frac{4(5)(1000)[1+1.5(6/8)]}{87(24)(8)(12)} = 0.21\, in. < 1/4\; in.\; allowed$$

2' Ø pile or shaft, embedded 8'-0" is adequate.

<u>Concrete Design</u>

Analysis by Broms method requires equilibrium at ultimate soil resistance (failure) and then application of a factor of safety to obtain allowable loads. This method does not have equations for maximum shear and moments in the pile at allowable or other lateral loads. Soil-pile interaction is highly non-linear, particularly at loads between Pa and PULT. To obtain moments and shears for a given applied load requires use of a non-linear computer program.

Conservatively, assume maximum shear and moments at allowable loads is the same as at PULT. This assumption does not necessarily result in excessive reinforcement . In most cases, nominal reinforcement develops the required shear and moment capacity.

For lateral loads less than 1/3 to 1/2 P_{ULT} , linear interpolation is permitted. Soil-pile interaction up to 1/3 to 1/2 Pult is approximately linear.

A) Moments

At P_{ULT} = 23.2 K, M_{maxp} = 214.8 FT-K

Assume, at P_a = 11.6 K, M_{maxp} = 214.8 FT-K

Linearly interpolate at P_{actual} = 5.0 K, M_{maxp} = 5 / 11.6 (214.8) = 92.6 K

Maximum ultimate moments = 1.275(92.6) = 118 Ft-K

8~ #6 O.K.

Ultimate axial load = (1.05)40 + 1.275(35) = 86.6 K

B) Shear

At $_{PULT}$ = 23.2 K, Maximum shear in pile = $4.5C_uDg$ = 4.5(2.4)(2)(4.46) = 96.3 K

Maximum shear is in the shaft and not bell (g = 4.46 FT)

For actual P = 5.0K, shear = (5 / 11.6)(96.3) = 41.5 K

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SAMPLE DESIGN 4: SHORT BELLED PIER IN CLAY

V_{ult} = 1.275(41.5) = 52.9 K

For 24" Ø pile, equivalent square = 21.27 IN

b_o= 21.27 IN

d = 18.27 IN

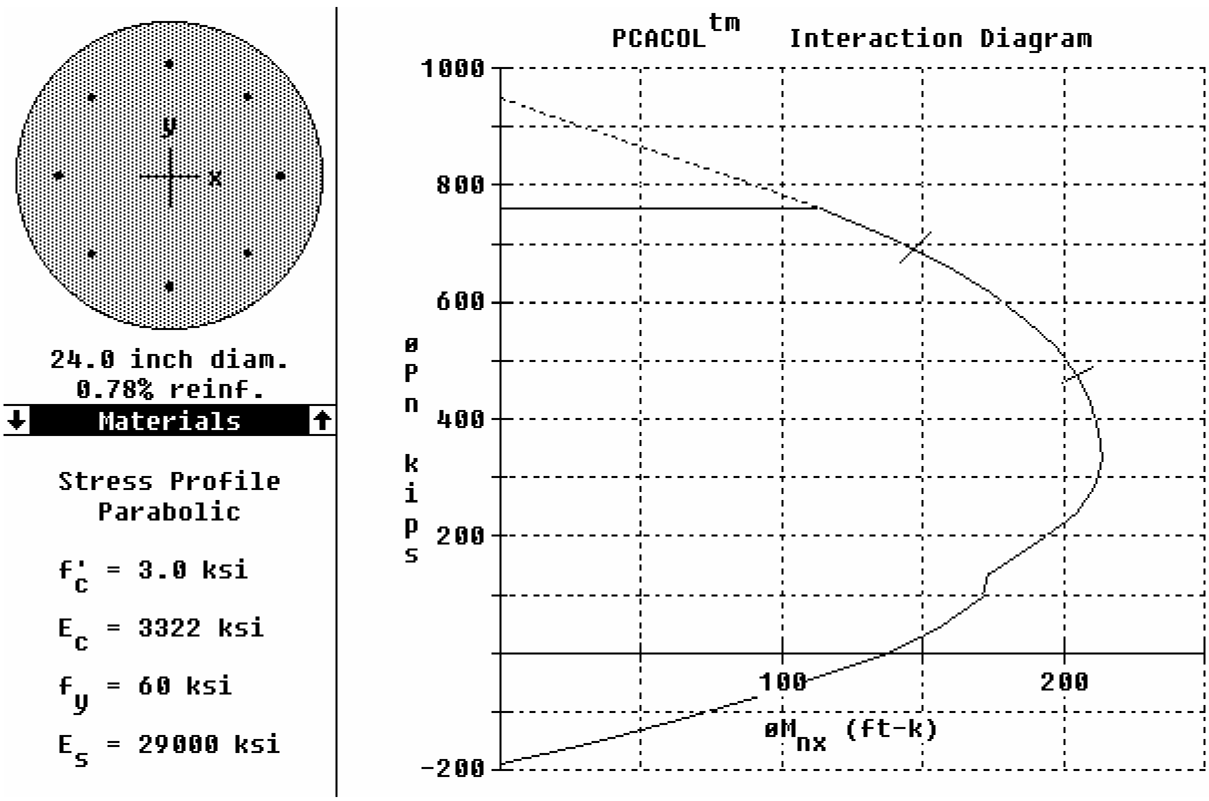
V_c = Ø(2) bod = 36.2 K

V_s = 52.9 - 36.2 = 16.73 K

S = $A_v f_y d$ / V_s = 2(0.11)(60)18.27 / 16.73 K = 14.41 IN

Use #3 ties @ 12IN

<u>Pile Section - Concrete Capacity</u>



Reinforcement:
- 8#6
- A_{st} = 3.52 in2
- Tied cc=3.00 in
- Spacing = 5.85 in
- I_x = 16286 in4
- = 0.00 in
- I_y = 16286 in4
- = 0.00 in

Also:
- ACI 318-89
- Reduction: phi c= 0.70
- phi b=0.90
- a=0.80
- ε_u = 0.003 in/in

Slenderness not considered x-axis